

EVALUATION OF CONTINUOUS CONTROLS FOR RADIAL MENUS

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ABSTRACT

This research focuses on a computer human interaction technique well suited toward ever-increasing display areas found on larger monitors, multiple monitors, and large interactive displays such as whiteboards. Specifically, a new type of interface control is explored; one that helps move the user interface from the edges of the screen to the location where the user is currently working. Existing techniques of moving the user interface to the user, such as gesturing or radial (pie) menus, allow the user to choose an item, but suffer from the inability to choose the value of an item. The introduction of continuous controls for radial menus allows for the selection of an item, such as with pie menus, but in addition allows for the selection of a value for that item. An example of this might be an audio menu where the items are volume, bass, treble, and balance, and the item and its associated value can be chosen in one action.

This research evaluates continuous radial menus through an empirical experiment. The results for the experiment show that (1) a visual mark to act as an indicator of the value about to be selected is not necessary, but may be desirable; (2) the direction of increasing value (top-to-bottom versus clockwise) does not effect performance, but perhaps clockwise is preferable; (3) the menu should have an outside radius of at least 100 pixels or approximately 2.5 cm, but

an unbounded menu is best; (4) a menu can support up to eight continuous control items without seriously degrading performance; and (5) the proposed menu is easy to learn.

DEDICATION

To my family.

My parents, Mary and Charles Fleck, my wife, Jean Herman, and my sons,
Charles and Joseph Fleck.

ACKNOWLEDGMENTS

Special thanks go to Dr. Susan Palmiter for her outstanding input and guidance, and likewise to my other committee members, Dr. Bo Sandén and Dr. Cynthia Calongne. Ben Straub also deserves special thanks for his work on the test software, and Brent Wilson for volunteering to be my coder for the experiment. And I never would have figured out the statistics without the superb help of Dr. Erik Nilson.

But most of all, none of this would have been possible without the support and understanding of my family. Working on a dissertation while simultaneously taking classes and working a full time job presented a constant struggle to allocate waking hours. My wife, Jean, stood by me the whole way, always showing consideration for my time. She's also an awesome editor, suffering through my many drafts and correcting my errant commas (any residual errors are most certainly from my changes after her review.) My sons, Charles and Joseph, patiently gave me space and waited for the day I was done. Yes, we can play board games now!

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CHAPTER I

INTRODUCTION

A computer user will typically arrange what is displayed on their screen so that the document being worked on is centered on the display, and menus and palettes are arranged around the border. Some items, such as the Macintosh® menu or dock, and the Windows® task bar, must always be located at the screen edge. As monitor prices have dropped, the use of larger monitors and multiple monitors has increased. With menu and palette User Interface (UI) elements at the screen edge, the amount of mouse movement needed to reach these items is increasing. This results in a slowing of performance, and distracts the user from the task at hand by forcing the user's gaze to the screen edge. To an extent, this challenge can be overcome by changing the ballistics profile of the mouse to increase the control/display (c/d)¹ ratio to allow the user to reach the edges more quickly, but at the cost of less control. With interactive pen displays and interactive whiteboards, the c/d ratio is fixed at 1:1 and as these devices grow in size, this further exacerbates the control problem. In addition, with these types

¹ The c/d ratio is the ratio of the physical distance traveled by the control (input device) and the physical distance traveled by the object on the display (such as the screen cursor.)

of devices, a large size can also lead to fatigue from the required extra movement.

Chapter II of this document reviews some of the work that has been done in an attempt to alleviate the problem of excessive mouse movement by bringing the interface closer to the user's gaze. Bringing the interface to the user's gaze consists of temporarily placing the controls on the screen where the user is focusing, as opposed to leaving them at the screen edge. Chapter II first looks at performance measurement techniques. As might be expected, performance drops as distance increases, which is the basic motivation for moving the interface closer to the user's gaze. Chapter II then looks in detail at radial menus, commonly referred to as pie menus. The chapter also presents crossing menus, a radial technique based on crossing a boundary to select an item, as opposed to clicking in a region. Marking menus are also discussed. Marking menus are a blend of a radial menu and a gesture. The marking menu is used initially during learning, and a migration to gesturing takes place as skill improves. As will be shown, the vast majority of radial menu work has focused on menu-like controls, where a command is selected. On the other hand very little work has been done to provide a continuous control, such as a slider.

Chapter III explores the development of a continuous control that could be incorporated into a radial menu. Use of such a control may lead to performance gains from reduced hand motion and a reduced need for the user to

shift their gaze. Incorporation of the proposed control into pie menus builds on the familiarity users have with pie menus instead of attempting to propose an entirely new control concept. Design considerations are given to the visual appearance and the nature of the interactions with the control. Although some aspects of the proposed control are well specified, questions remain as to some of the parameters and the usability of the control. Four of these questions were chosen for further examination. These questions are:

- Does a visual mark improve performance?
- Which orientation of increasing value is best?
- How many continuous menu items can a menu contain?
- Is a continuous pie menu easy to learn?

These outstanding questions were then formalized in the form of the following hypotheses.

- Hypothesis 1: A visual mark will yield a faster response time and lower error rate than no visual mark.
- Hypothesis 2: A top-to-bottom orientation will yield a faster response time and a lower error rate than a clockwise orientation.
- Hypothesis 3: Collectively, users will tend toward a radius that is neither too close to the center, nor too far from the center. This will happen with a relatively small standard deviation.

- Hypothesis 4: Menus that subtend a larger angle will yield a faster response time and a lower error rate than menus that subtend a smaller angle.
- Hypothesis 5: Selection time will quickly decrease then level off with use.

This chapter also describes the experiment that was designed and run to test the hypotheses. The experiment controlled the presence or absence of an indicating mark, increasing values in both clockwise and top-to-bottom orientations, and four, six, or eight items in the menu. Other experimental variables were also controlled to reduce variability, such as which menu items were chosen, and what value was chosen. Performance time, errors, and the distance the user's hand moved from initial activation were measured via a software program developed for the experiment.

Chapter IV describes the detailed results of the experiment. The inclusion of a visual mark and the direction of increasing value, were found to be insignificant. However, the number of items was found to be significant. It was also found that users in general tended to prefer a particular selection distance from the center. The new control was also found to be easy to learn.

In Chapter V, the results were used to develop guidelines for practical implementation of the control. Recommendations for the visual layout and minimum diameter are provided. Chapter V also provides a summary of the work, and discusses open questions for possible future research.

CHAPTER II

REVIEW OF LITERATURE

In this chapter, the motivation behind radial selection techniques will be examined. Following this, three primary radial selections techniques will be explored. These include pie menus, crossing menus, and marking menus. Lastly a few other techniques, including continuous controls, will be explored.

2.1 GUI Performance Fundamentals

The inspiration for radial selection stems from Fitts' Law [1]. Fitts did early research on human movement for time and motion studies for industrial purposes; studies which have direct application to input devices. His experiments with tapping a stylus on rectangular electrodes led to formulae for determining time and difficulty of selecting targets as a function of target size and distance to the target. Figure 2.1 shows his experimental setup.

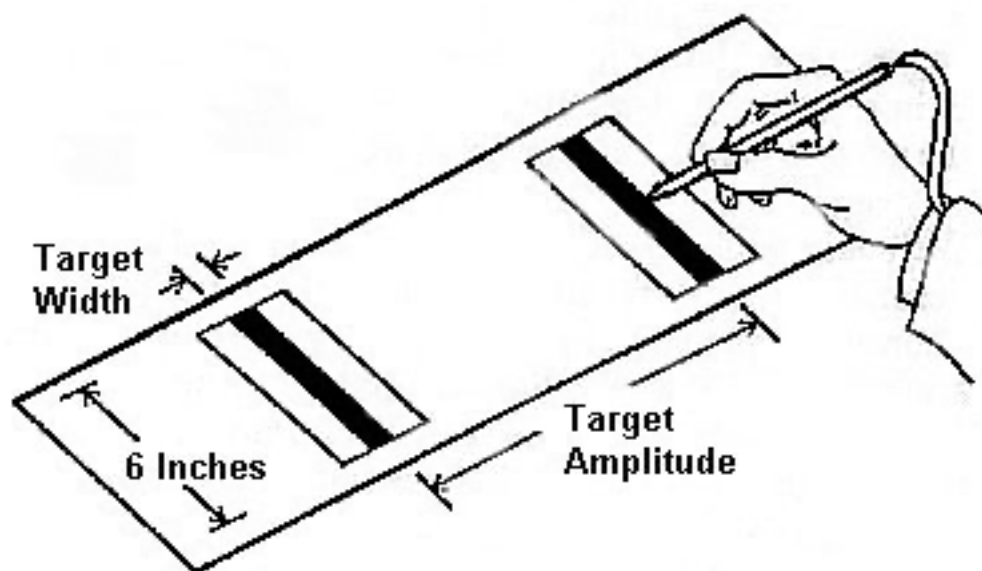


Figure 2.1 Fitts' tapping experiment setup [1]

The basic conclusions of his study were that the further away the target is, or the narrower the target is, the more difficult the target is to select, and the higher the likelihood of missing the target. Fitts based his work on Shannon's Information Theory, which resulted in his difficulty measurement (I_D or Index of Difficulty) being measured in bits, and his performance measurement (I_P , or Index of Performance) being measured in bits per second. [2] These formulae have direct application to selecting icons, buttons, and menus in today's UI's, and are widely used when investigating performance in UI's. However, Fitts' experiments were inherently one-dimensional, whereas current Graphical User Interfaces (GUI's) are two-dimensional. MacKenzie examined Fitts' Law as it applies to UI's and proposed three improvements. [3] First, he demonstrates that

the Shannon formulation of Fitts' Law is slightly better than the original, as it never produces negative results and has a slightly better fit. The Shannon formulation is shown in equation 2.1.

$$ID = \log_2\left(\frac{A}{W} + 1\right) \quad (2.1)$$

Second, MacKenzie derives an interpretation of the width parameter that can be applied to both one and two-dimensional tasks. Third, he demonstrates how to normalize performance based on errors. At this point it might be obvious that Fitts' Law is the core reason for using radial selection techniques. By displaying the selections at the current screen pointer location, distance is always minimized, leading to improved performance.

This is all well and good for a simple pointing task. However, menus, including radial menus, are often extended through some means of hierarchy. In these cases, a path must be followed. Accot and Zhai performed several experiments of various path complexities and derived formulae for path following known as the Steering Law. [4] Accot and Zhai demonstrated the use of the Steering Law by applying their formula to a hierarchical menu. They tested their formula with various input devices [5] and also tested the Steering Law on paths of different sizes. [6] When they did this, they found that there is some scale effect, probably due to different joints being used and limits in motor precision. Balakrishnan and MacKenzie investigated the performance of various human joints. [7] They examined different finger, wrist, and arm movements and

concluded that fingers underperform other limb segments in pointing tasks. Therefore there is some scientific basis to the variations that Accot observed.

Current literature predominately uses Fitts' Law and Accot's Steering Law to evaluate user interfaces. However, there are other models. The Complexity of Interaction Sequences Model breaks down interaction sequences into steps and predicts the performance as the sum of the individual steps. [8] The authors applied their model to fixed palettes, bimanual palettes, and toolglasses, and their experiments supported their predictions.

2.1 Pie Menus

The earliest form of a radial selection technique is a pie menu. Although there are some earlier references, such as the movable menus by Wiseman et al. [9], the seminal paper on pie menus is the article by Hopkins in Dr. Dobb's Journal. [10] Hopkins provides an excellent summary of their workings and includes references to studies done by others before the article was written. One of the studies Hopkins cited was a comparison of pie menus to linear menus by Callahan, which determined that pie menus are indeed faster and less error prone than linear menus. [11] Figure 2.2 shows an example of a pie menu.



Figure 2.2 Example pie menu [12]

Pie menus can be textual, iconic, or a combination of both. In some implementations, only the text is shown with no background or borders. Clicking in a pie segment makes a selection. The number of choices can be extended through a hierarchy, where a second level menu is displayed at the location where the first menu item was chosen. The menu is only displayed when activated, which is usually done with a keyboard key or a right mouse click. In a process called “mouse-ahead”, the display of the menu can be delayed for a moment. In cases where the user has memorized the positions, with mouse-ahead they can make a selection without the menu displaying, avoiding distractions and potential delays on slower systems.

There have been numerous variations on the pie shape. For example, Lyons et al. experimented with various layouts such as spiral, concentric circles, circles with a hierarchy, and, finally, oval layouts. The oval layout allows the menu text to remain horizontal with lots of space for the item text while still

using a radial selection from the center circle. Second and deeper levels are drawn at the same location. An inward arrow in the center allows for "go-down" (back down a level). Figure 2.3 shows his resulting menu. [13]

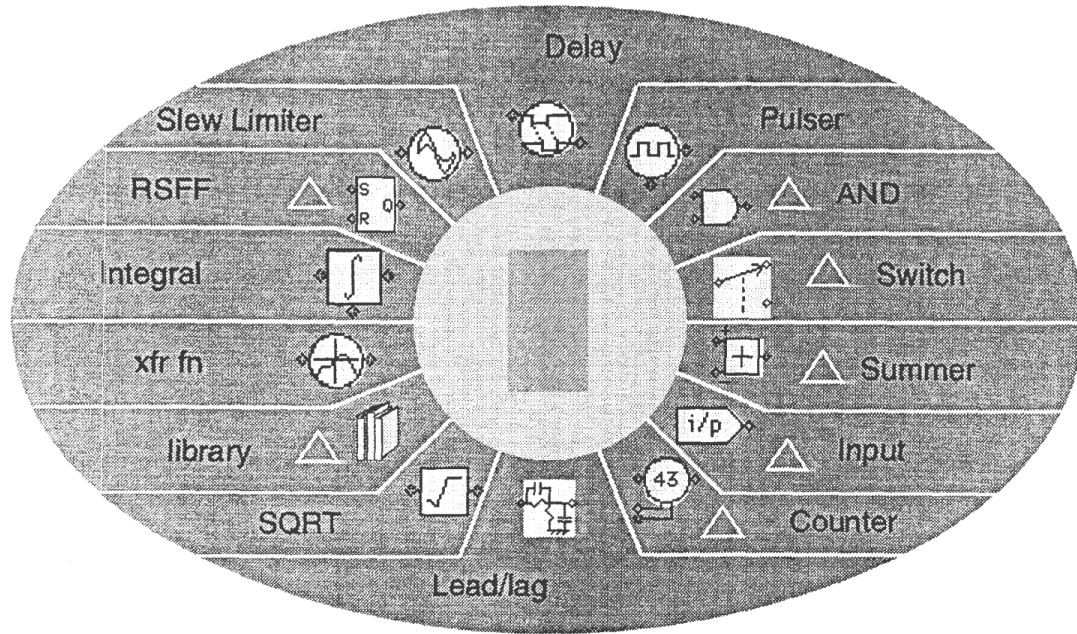


Figure 2.3 Oval variant of a pie menu [13]

A search of published patents also produces a number of variations of menu shapes and arrangements. In Anderson et al., the menu is presented in a curved band, instead of an incomplete circle. [14] This makes the items easy to reach with a sweep of the hand. With the curved band, hierarchical items are presented inward toward the hand. This is in contrast to the patent by Abrams et al. where previously selected items in the hierarchy are pushed toward the center of the menu and new items are displayed at the perimeter. [15] The patent by

Atkinson combines a pie menu with a linear menu. [16] In Atkinson's implementation, a portion of the menu, for example the bottom half, or the lower right quadrant, is a linear menu. An example is shown in Figure 2.4. This variant allows for displaying a greater number of items than can usually be accommodated in the pie menu alone. The less frequently used items are placed in the linear menu. Note that the linear portion is quicker to use than standard linear menus, since the linear portion of the menu is always close to the current screen pointer instead of attached to a window or at the top of the screen.

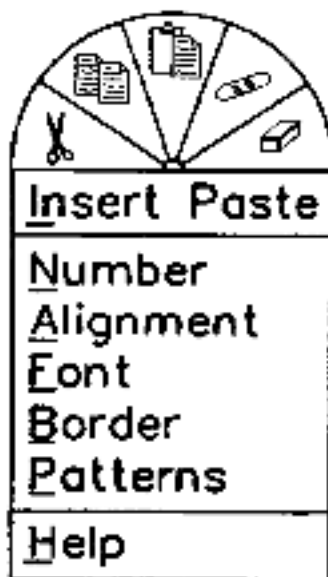


Figure 2.4 Atkinson's combined pie and linear menu [16]

The implementation in the Driskell patent stretches the oval of the Lyons implementation into a rectangle, and aligns the options into two columns. [17] The center still allows radial selection, but according to the patent, aligning the

items vertically makes searching through the options in the list faster, because more items fit in the eyes' field of view. Driskell's implementation can be seen in Figure 2.5.

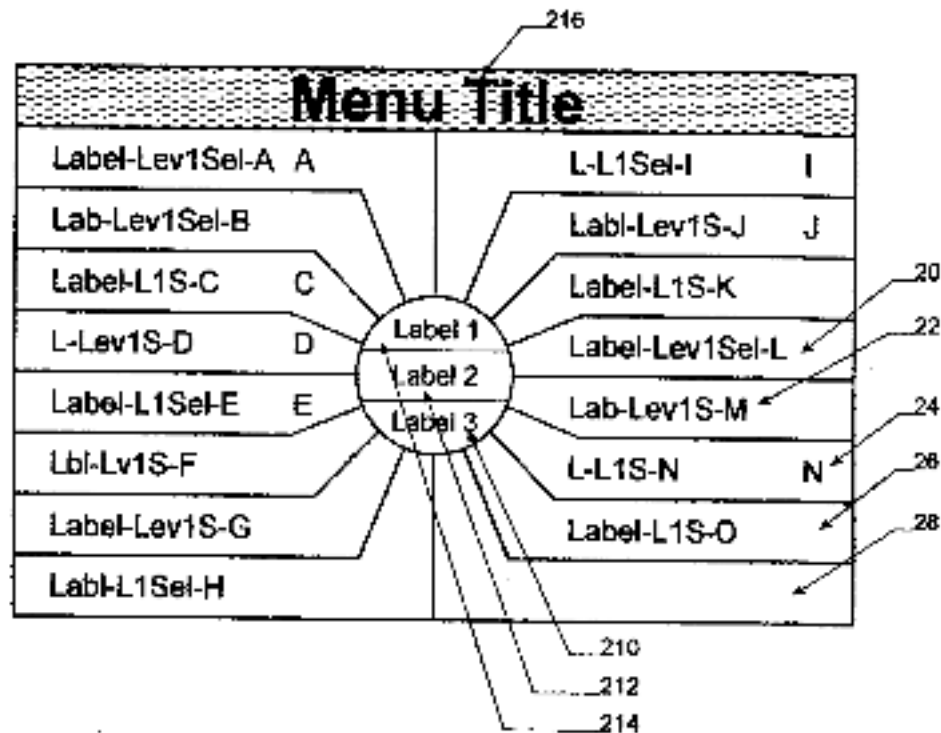


Figure 2.5 Driskell's pie menu variation [17]

In the patent by Selker, the menu consists of concentric circles where the inner items are used most frequently and the outer ones less frequently. [18] He also discusses an implementation where the outer circles are hierarchies built on the inner selections (as opposed to displaying a new pie at the selected location). He also discloses making the menus dynamic, where more frequently used items, or favorite items, are given a preferential location toward the middle. Vayda and Weyl arrange their menus in concentric rings. [19] Each ring has a

title area located on a tab that extends into an adjacent ring. The tabs are staggered to avoid overlap.

The literature is also rich in other variations of pie menus, or pie menus combined with other interaction techniques. For example, Hinckley et al. combine a gesture with a pie menu. Objects are selected via a lasso gesture, or in another implementation, by crossing over them. [20] At the end of the gesture, a pigtail is made to indicate that the menu should be displayed to choose an action (such as cut, copy, move, etc.). This is shown in Figure 2.6. The pigtail itself can indicate a direction and make a menu selection from the pie menus without the need for extra strokes. The menu selection can even be made without the menu being displayed by using mouse-ahead.



Figure 2.6 Pie menu combined with a selection gesture [20]

The text input method by Venolia and Neiberg starts with a palette of character groups. [21] The user then chooses a character in the group by drawing a vector starting in the initial group. The pie and vector are drawn with an offset

from the current position of the user's hand to prevent the hand from occluding the menu. There is a training mode where the menu is always displayed. By the end of three sessions, the need for further training was less than half, and essentially totally gone by nine sessions.

Kurtenbach et al. developed a method where menu strips are displayed around the screen cursor. [22] As can be seen in Figure 2.7, this method allows a large number of selections to be displayed at once.

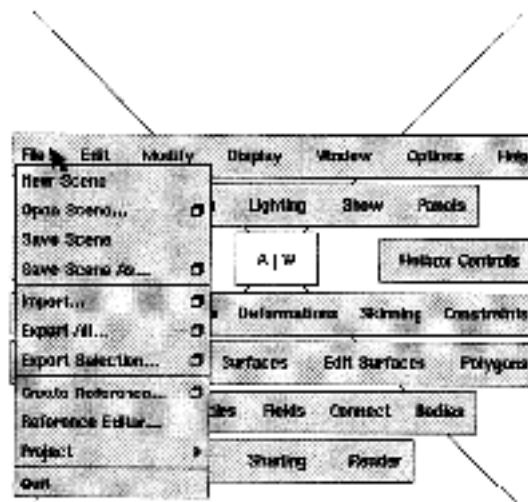


Figure 2.7 Radially displayed linear menus [22]

In the literature, there is one mention of continuous selection in a pie menu. In a paper by Hopkins written in 1987 (but not appearing to have been published until it was placed on the world wide web in 2005), Hopkins suggests selecting a value based on the distance from the center of the selection point. [23]

There are no known implementations that use this technique. Hopkins' suggested menu is shown in Figure 2.8.

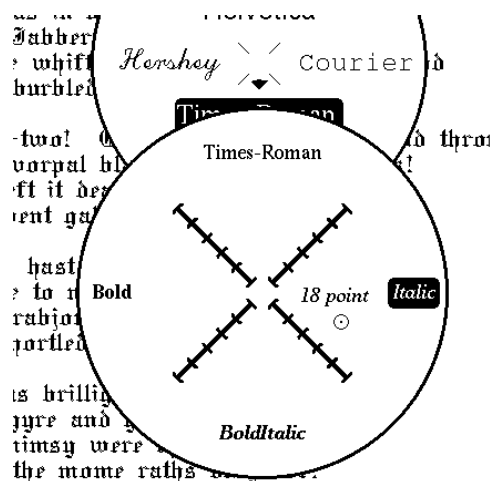


Figure 2.8 Distance based selection [23]

2.2 Crossing Menus

In crossing based interfaces, users select items not by clicking, but by merely moving over them. Accot and Zhai studied crossing interfaces and demonstrated that they follow Fitts' Law. [24] In the course of doing their Steering Law studies, they showed that crossing interfaces are equal to or better than clicking based interfaces when it comes to accuracy and performance. [4, 5] Winograd and Guimbretière demonstrated how a crossing interface works particularly well with large displays using a laser pointer. [25] Cockburn and Firth studied interactions with small targets and concluded that goal crossing was more efficient than expanding targets or sticky targets, while still

conforming to Fitts' Law. [26] Lapizco-Encinas and Rodríguez applied goal crossing to an editing program called "CrossEd". [27] They provided an excellent mapping between standard controls, such as radio buttons and checkboxes, and crossing based equivalents. Apitz and Guimbretière applied crossing techniques to a drawing application called "CrossY" that used crossing concepts in its interface. [28] They claimed that the interface encourages more fluid interactions, which may result in more efficient and natural feeling pen interfaces.

When crossing methods are applied to radial menus, the user is only required to cross into a sector, not cross and click to select. Guimbretière and Winograd used this method in the "FlowMenu". [29] In a FlowMenu, the user clicks in the middle, and then makes gesture-like selections. Figure 2.9 shows selecting a 100% zoom-factor in a FlowMenu. In Figure 2.9a, the user is moving toward the Item menu. Once inside the Item menu, the other first level menus are grayed out and the second level menus are displayed. In Figure 2.9b, the user is crossing from Zoom back to the center to select the Zoom function. At this point the items change to zoom-factors, as seen in Figure 2.9c. (Note that the path followed is not drawn on screen, but is shown in the figure for explanatory purposes.) There are several advantages to this method. One advantage is that the menus don't "drift". "Drift" is when the menus "walk" across the screen as the hierarchies are displayed at new locations. By not drifting, the menus can be closer to the screen edge without subsequent menus falling off the edge of the

screen. Another advantage is that the number of items that can be selected can be increased by using crossings across the radial arms in addition to crossings to and from the middle. Also, the direction of crossing an arm can be made significant, giving even more options.

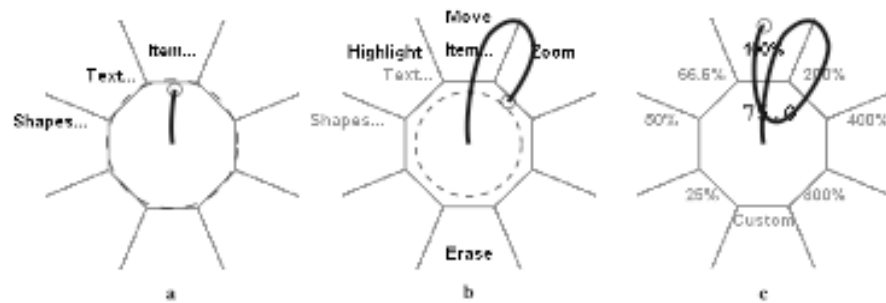


Figure 2.9 Interactions with FlowMenu [29]

Bederson used FlowMenus as an example of an interface that provides an optimal experience. [30] To be optimal, the interface must change to reflect the user's skill level as they progress from novice to intermediate to expert. With FlowMenus, novices are given visual guidance, but by the time a user reaches the expert level they will have learned gestures corresponding to the commands and will no longer need the visual cues. Pook et al. proposed control menus, which are very similar to flow menus. [31] In a control menu the action is triggered once a distance threshold has been exceeded. In some cases the action that is triggered is a continuous action, such as zooming or scrolling. The action continues until

the click is released. Although the method allows for a continuous value, it does not allow for a selection of an absolute value.

2.3 Marking Menus

Another form of radial selection is marking menus described by Kurtenbach and Buxton. They define a marking interface as “interactions where the pointing device leaves an ‘ink trail’ on the display similar to writing with a pen.” [32] They used GEdit, a prototype graphics editor, as a testing platform for marking techniques. One of the techniques they tried was a combination of pie menus and marks. The menus themselves resemble pie menus with mouse-ahead, but are characterized by the selection method. With marking menus, selections are made by clicking in the center and drawing. When combined with mouse-ahead, the drawing devolves into a gesture. Marking menus also provide an optimal user experience. They are menu-based for the novice to promote learning, and eventually gesture-based for the expert. Kurtenbach did extensive experiments with marking menus, studying the number of menu items (from 4 to 12), the type of interface device used (mouse, pen, or trackball), and whether the menus and marks were displayed or not. [33] He found that a stylus was the best choice for marking menus, and that always displaying the menu not only resulted in lower response times, but also fewer errors. He also discovered that even numbers of menus were superior to odd numbers. Kurtenbach then extensively studied the marking menu concept as his PhD thesis. [34] He

investigated both hierarchical and non-hierarchical marking menus. He gave detailed consideration to the interactions with marking menus, such as how to abort, how to back up a level, and how to avoid ambiguities, such as distinguishing two marks in the same direction from a single mark. He discovered that the users considered the task a selection task, and not in terms of marking or drawing. This became obvious when users would draw lines that weren't straight because they changed their minds mid-mark while making a selection. As might be expected, the more items shown at a level, or the more levels (even with fewer items), increased both errors and response time. Eight items in a level seemed to be optimal. Tapia and Kurtenbach then further refined marking menus. [35] They displayed only the menu text without the wedges. This allowed more of the underlying document to be observed. They standardized on eight items with a compass star in the middle. They used "idealized" marks (the drawn lines were "cleaned up", or straightened). This made the final mark resemble the desired gesture, and gave users confidence about their marks. Kurtenbach further studied learning and use of marking menus. [36] Figure 2.10 shows the improved implementation.

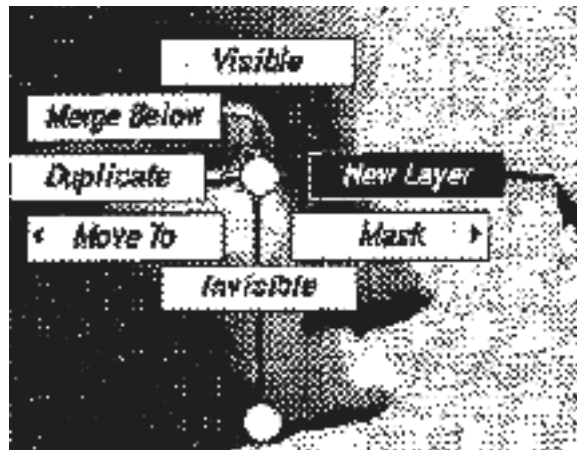


Figure 2.10 Improved marking menu [36]

Kurtenbach found that marking menus are indeed more efficient than linear menus, sometimes by as much as 3.5 times faster. However, he discovered that users would drop back to using linear menus instead of marks after a lay-off period, demonstrating the necessity of having the menus always available, even to expert users. The marks were successful in "an appropriate setting", specifically when there were a few frequently used marks that required a screen location as a parameter. They were not as successful when the list was dynamic or changed frequently. Kurtenbach disclosed a method to extend the number of items in a marking menu in his 1997 patent. [37] By replacing one of the items in the marking menu, such as the "south" item, with a linear menu, a larger number of selections could be included in the menu. Ideally, less frequently used items would be in this list since having too many items present could be expected to result in lower performance due to searching and movement times.

Kurtenbach confirmed that errors went up with the number of levels in a marking menu. Zhao and Balakrishnan tried to overcome this by using a series of simple marks instead of a single complex one. [38] Drawing individual marks took less screen space and resulted in somewhat faster response times. With individual marks, the errors did not accumulate as rapidly as they did with compound marks. This resulted in more accurate results and allowed for deeper hierarchies. In Mackay's explorations of floating palettes, marking menus, and toolglasses, she discovered that some interaction techniques encourage the user to do as much of the current activity as possible before switching to another activity. [39] Of all of the techniques studied, marking menus were found to encourage the most activity switching. Marking menus were best used with activities where several different commands are executed on a single object before moving on to another object. This is in contrast to repetitive commands, such as repeated copy commands, where floating palettes excelled.

2.4 Other Techniques, Including Continuous

There are a few other interaction techniques of note. Some are variations of the techniques already described, and some are unique. With few exceptions, radial selection techniques are used to choose menu items. These are almost always command oriented, such as cut, copy, paste, or change font or style. McGuffin et al. proposed a technique they call FaST Slider that allows continuous values to be set. [40] The technique starts with a marking menu. Once

an item is chosen from the menu, the menu is replaced with a slider control. The slider follows the screen pointer until the user performs a click, which pins the control at the location of the click. The slider can then be adjusted until dismissed. When the user moves off of the slider, additional buttons are displayed next to the slider that allows dismissal, cancellation, or incrementing or decrementing the value. These steps can be seen in Figure 2.11.

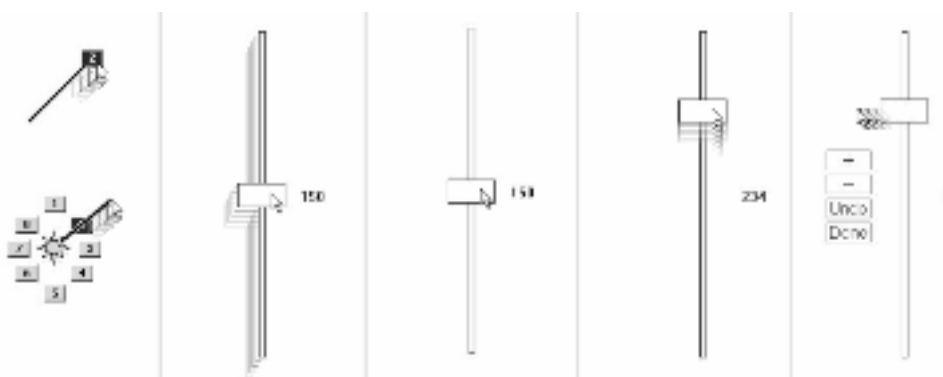


Figure 2.11 FaST Slider interaction [40]

Smith and Schraefel introduced a radial scroll tool. [41] As shown in Figure 2.12, a series of radial lines are displayed starting at the point of activation as a cue. A clockwise gesture over the radial rays "advances" the document. The opposite is true of counterclockwise gestures. In the final implementation, the center point was offset 20 pixels below the starting point to avoid the initial movement out of the center. This placement of the center point was found to be more efficient for scrolling short distances, but traditional scroll controls were

better for long distances. Since the control only increments and decrements, it is not suitable for picking an absolute value.



Figure 2.12 Radial Scroll Tool [41]

Microsoft's Vista operating system includes a "pen flicks" feature. [42] With this feature, quick strokes along compass directions execute commands. This is similar to marking menus without the menu, but unlike radial techniques, does not act upon the object under the screen pointer when executed. This feature behaves like a gesture technique, such as the gestures that are part of the Mozilla web browser. [43] Indeed, gestures in general have the benefit of being executed at the screen pointer location, and so have the benefit of Fitts' Law. However, they do not allow the user to choose a continuous value. Further analysis of gestures is beyond the scope of this research.

Radial selection techniques can boost performance by bringing interactions in the user interface closer to the user's working location. So far, adoption of these techniques has been slow despite the benefits. Hopkins has put forth several reasons for the slow adoption of pie menus, including the lack of a pie menu as a standard built-in control in modern operating systems. [44] However, as monitor sizes grow, these techniques will become even more

valuable. To date, most applications of radial selection techniques have involved command execution, such as those done with linear menus. The proposed control extends pie menus with the ability to select from a range of continuous values. The method is more efficient than previous attempts as it allows selection of a control and a value in a single click.

CHAPTER III

RESEARCH METHOD

3.1 Menu Design

The goal of this research is to design, evaluate, and refine an improved continuous control menu item. The proposed control is intended to have all of the advantages of radial menus while also providing the ability to select a continuous value. This chapter sets forth an initial design of the control, including rationale behind that design. While researching the literature, and during the design process, questions were raised about design features. This chapter then describes an experiment that was then performed to evaluate the control and answer the questions raised during design of the control. In the following chapters, the experimental results will be presented and those results will be used to recommend refinements and follow-up research.

The design of the menu is intended to extend a basic pie menu with the addition of pie item segments that will allow for the selection of continuous values. The intention is to use these items in place of the usual menu/command items so as to allow users the ability to select from a range of values, something

not currently possible in a pie menu. For example, by using these new menu items a user could choose a line thickness or a sound volume. Fitting options like these into an existing pie menu however imposes constraints on the overall menu. The most obvious constraint is the size of the menu. The continuous menu item may require additional space due to the reduced target size. The issue of target size will be discussed in detail after the visual design is described.

Fitting numerous options within the visual layout of a pie menu is constrained by the outline of the proposed control. It's a wedge of a pie. This constraint can be relaxed by leaving open the outside boundary of the item. Although historically pie menus have tended to be bounded, recent implementations have incorporated both bounded and unbounded menu items. For example, in the Second Life[®] virtual world application [45] the menus are both bounded and unbounded. In this program on the Windows[®] operating system, menus are invoked with a right click. If the right button is held after the click, the menus are unbounded and can be selected by moving over a menu item and releasing the right click over the item. This has the potential of being much more efficient, since the unbounded menu can be much larger, and a selection takes place with a single click and release. On the other hand, if the user releases the right click in the middle of the menu (which is where they originally did the right click), a more traditional pie menu with boundaries is presented. In this case, the menu items are selected with a left click. Although there are

obvious benefits to the unbounded menu, it may not be immediately clear to users how to use an unbounded menu, since most pie menus don't offer a right click select option similar to Second Life®. Canceling the action with an unbounded menu is less intuitive since the center must be used to cancel. With a bounded menu, a click outside the area of the menu acts as a cancel. Since there is no "outside" of the menu for an unbounded menu, the center must be reserved for a cancel.

Consideration was given to the visual design of the menu. Three types of menus were examined before selecting the final design. The first option was a menu employing a simple opaque circle such as shown in Figure 2.2. The problem with using a pie menu with an opaque circle design is that during operation, opaque menus obscure the object after the menu is activated, requiring the user to remember visual details about the object. A second choice was to use a text-only pie menu, much like the menus in the marking menus in Figure 2.10. However, text-only menus suffer from a serious deficit. With text-only menus the boundaries of the wedges are not visible. The user is left unsure about where the target area is for selecting an item. Is the target area a virtual wedge encompassing the text, or is it only the area of the text? This is further complicated by the fact that longer text on some items often overlaps into the virtual wedge of adjacent items.

The final menu design considered was one using transparent circles such as the ones used in the program Second Life®. The transparent menus in Second Life® allow the underlying objects to be seen, while still presenting the pie menu. Figure 3.1 shows a Second Life® menu in its unbounded state. The reduced transparency in the middle allows for the menu text to be placed on top of objects and still be readable even on light backgrounds. The Second Life® menu gets more transparent further from the middle to eliminate the concept of a boundary. Because of these benefits, a visual design similar to the unbounded menus in Second Life® was chosen.

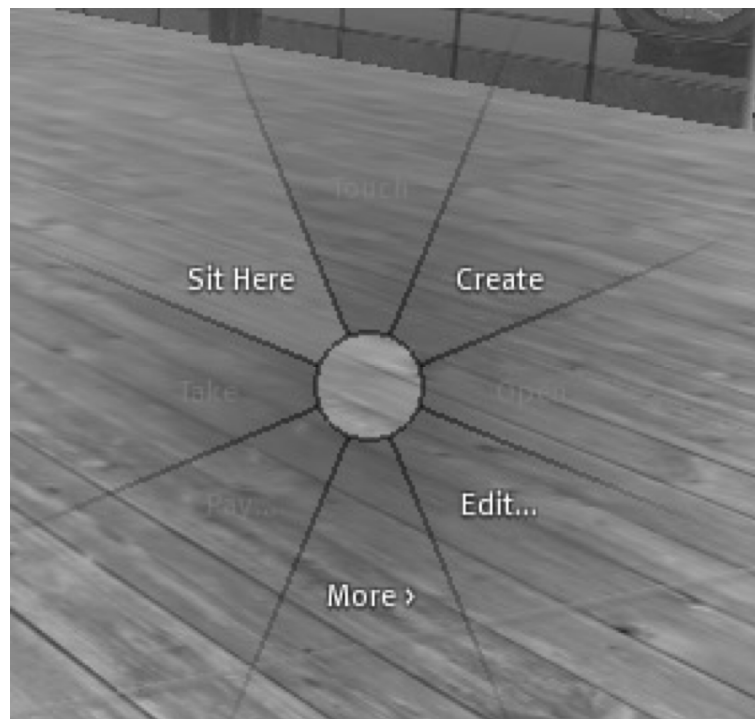


Figure 3.1 Example Unbounded Menu in Second Life® [45]

Once the visual design was chosen, attention was next given to the method for the selection of a value. The continuous pie control requires not only selecting one of the wedges to choose an item to control, but also choosing a value for the item by selecting the appropriate location within the particular wedge. There are two possible dimensions that can be used to specify a value based on the location in the wedge. The value can be chosen based on the distance from the center of the wedge; or it can be based on the angle within the wedge. The targets for selecting a single value using these two techniques, distance and angle methods, are shown in Figure 3.2.

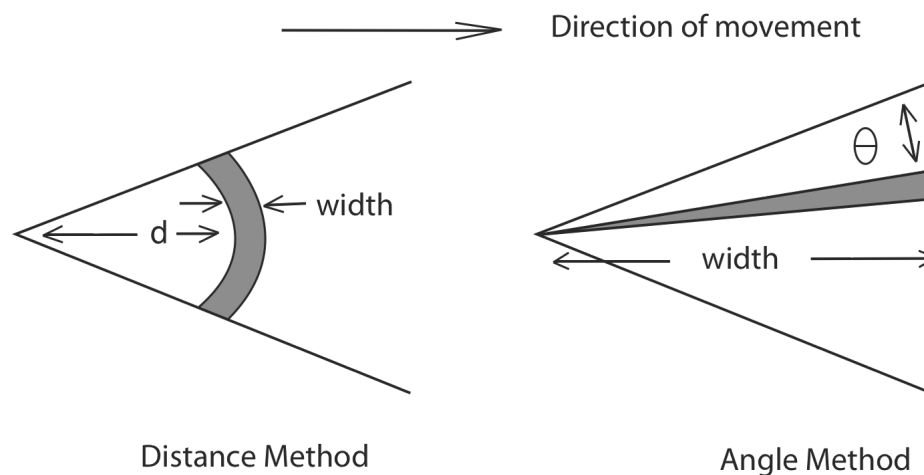


Figure 3.2 Possible targets for selecting a value

When evaluating the better method for value selection, the dimension of interest is target width (the value needed to compute the Fitts' Law time estimate). According to Fitts' Law, the larger the target width, the easier it is for a

user to hit the desired target; the height of the target area is much less important. In Figure 3.2 the gray area in each of the methods represents the target width for one particular value. Note that in the distance method the target width remains the same size, a narrow band, as one follows the direction of the user's movement. Selection of small numeric values occurs near the center, and these values have a very small overall target area due to the narrow width and the smallest height in that position. Larger values have a somewhat larger overall target area, with the largest value occurring at the menu boundary (this value's target area has the same narrow band width, but a larger height). As can be seen on the left in Figure 3.2, target widths remain narrow in the direction of movement, which according to Fitts' Law will result in poor performance and a high error rate. [1] Additionally, unbounded wedges become impossible to use with this approach, because the highest value is at the outer diameter of the menu, which doesn't exist in an unbounded menu.

In contrast, in the angle method the target width is as large as possible, since the width is always the same as the radius of the menu item. Furthermore the overall target selection area grows as the height increases as the user's directional movement travels away from the center of the pie menu. Clearly, the angle method of selecting values is superior over the distance method because of the larger width. With the angle method, all values have the same target area. Since the target is wide in the direction of movement, as shown on the right in

Figure 3.2, one would expect performance to be better and errors lower than with the distance method. The angle method has the added benefit of allowing unbounded selections to be made far from the center. With unbounded menus the target width in the direction of movement is very large (it is bounded by the screen edge, and the screen edge will often be very far away). The target height also becomes larger. The method of choosing a value based on an angle rather than on distance was chosen for the proposed design because of the superior benefits.

Normally the item about to be selected in a pie menu is obvious because it is the item under the screen cursor. This is sometimes reinforced by highlighting the item. With the continuous pie menu, the visual presentation needs to include some feedback showing the selected value. Figure 3.3 shows the proposed visual layouts with numeric feedback, and with and without an optional visual mark.



Figure 3.3 Visual layout with and without feedback

With these layouts, the minimum and maximum values attainable are shown along the respective edges. For feedback purposes, the value of the

current location is shown in a box in the middle and is continuously updated as the user moves about the control. The current value indicator is only shown in the menu item that the screen cursor is currently over. For consistency and simplicity in the study, numeric percentage values were used. In an actual implementation, other types of values could be used, such as a number of pixels for a line width, or 0 to 10 for a volume control. The right hand figure shows a layout that includes a mark to indicate the current angle. In this alternative, a line would “rubber-band” from the center of the menu to the location of the screen pointer. This would only be shown while the screen pointer was over the menu item and would disappear once a value was selected. The purpose of the mark is to help reinforce the fact that an angle, and not the distance, is the factor being controlled. The value of using the mark as a feedback mechanism was an open question studied in the experiment.

One more visual detail is the orientation of the high and low values within wedges with different orientations. Two possibilities were considered. Since a page is read top to bottom, and the orientation of the text is arranged to accommodate this, one option is to increase the value linearly from top to bottom. This is shown on the left in Figure 3.4. Another possibility is to increase numbers radially, such as in a clockwise direction. This is shown on the right in Figure 3.4. Both orientations of increasing values were studied to determine the optimal direction.

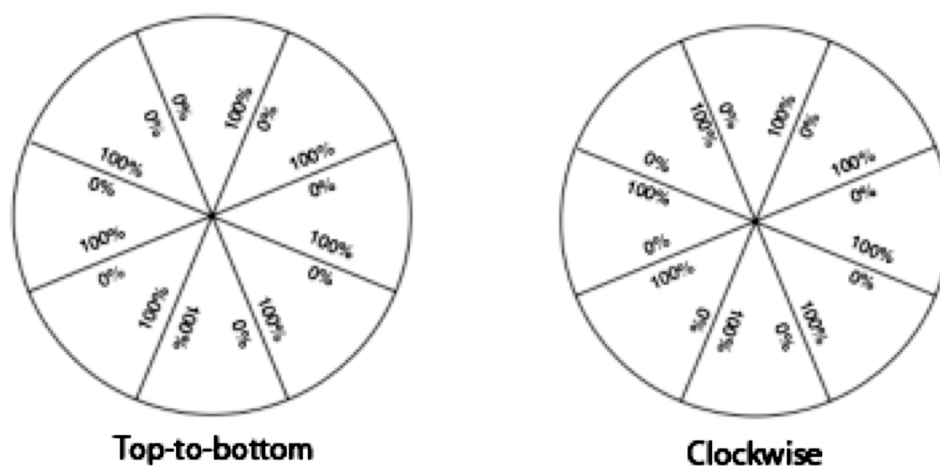


Figure 3.4 Possible value orientations

Now that the basic layout and interaction are defined, further investigation of the size of an item is in order. Figure 3.2 is a bit of an oversimplification, the details of which will be addressed here. There are two dimensions that need to be considered when investigating the size of a pie menu item. These dimensions are the overall diameter of the menu, and the arc subtended by an individual menu item. Pie menus typically have eight to twelve items. As will be seen, more than eight items is only feasible for continuous menus with very large diameters. Although there are no known studies of even versus odd numbers of items in pie menus, odd numbers of items are rarely used. Additionally, research on marking menus, which have strong similarities to pie menus, showed that an odd number of items was found to have equal or lower performance than the next higher even number. [34] Therefore, menus

with the even numbers of four, six, and eight items were considered here. These correspond to angles subtended by a menu item of 90° , 60° , and 45° respectively.

The smallest target possible is a single pixel. If the range of numbers is 0 to 100 in increments of 5, a total of 21 pixels are needed to achieve each value. This dictates the minimum arc of no less than 21 pixels in length. If an attempt is made to select any given value at a point closer to the middle than this minimum arc, not all values will be attainable. The distance from the middle to an arc of 21 pixels is shown as distance d_{21} in Figure 3.5. This distance will vary with the angle subtended by the menu item.

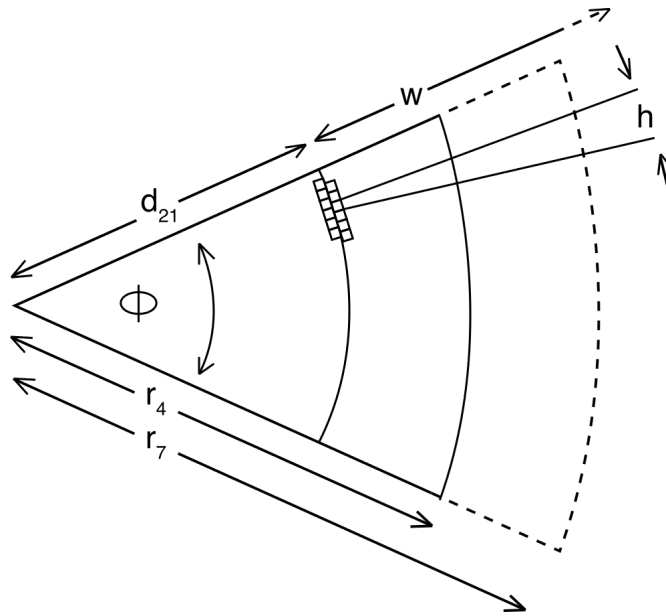


Figure 3.5 Details of target parameters

The parameters in Figure 3.5 are as follows:

d_{21} – The distance from the center of the menu to the arc where every value in a 5% control (21 pixels) can be selected. d_{101} would be the distance for a 1% control (101 pixels)

Φ – the angle subtended by the control. This is 90° for a menu with 4 items, 60° for a menu with 6 items, or 45° for a menu with 8 items

r_4 – radius of a 4 cm diameter menu

r_7 – radius of a 7 cm diameter menu

w – target width for a single value

h – maximum target height for a single value (the target is one pixel at d_{21} and gets larger until it reaches h at the maximum radius)

Table 3.1 d_{21} and d_{101} Minimum Distance at 100 DPI

Angle (# of items)	Minimum Distance	
	d_{21}	d_{101}
45° (8)	0.68 cm	3.27 cm
60° (6)	0.51 cm	2.45 cm
90° (4)	0.34 cm	1.63 cm

The formula for determining target width, w , is as follows:

$$w = \frac{\text{diameter}}{2} - \text{mimum distance} \quad (3.1)$$

Table 3.1 above details the minimum distance for d_{21} for various item angles on a display screen with 100 Dots Per Inch (DPI) resolution. As can be determined using the above table and formula, a menu item from a 4 cm diameter pie menu with 8 items would have a target width (w) of 1.32 cm for a 5% resolution control. This would be calculated as follows: a 4 cm diameter menu has a 2 cm radius (diameter/2); using the table, d_{21} for 45° (8 items) is =

0.68 cm minimum distance; a 2 cm radius – 0.68 cm minimum distance = 1.32 cm target width, w .

Similarly, using Table 3.1, the minimum distance for a 101-pixel arc, which can achieve a resolution of 1% cannot be accomplished with less than a 6.54 cm diameter menu with an 8-item menu. This would be calculated as follows: a 6.54 cm diameter menu has a 3.27 cm radius (diameter/2); 3.27 cm radius – 3.27 cm minimum distance for $d_{101} = 0$ width.

Table 3.2 Target height, h , for 5% resolution at 100 DPI

h in Pixels	Menu Diameter						
Angle	4 cm	5 cm	6 cm	7 cm	8 cm	9 cm	10 cm
45°	2.94	3.68	4.42	5.15	5.89	6.63	7.36
60°	3.93	4.91	5.89	6.87	7.85	8.83	9.82
90°	5.89	7.36	8.83	10.31	11.78	13.25	14.72

Table 3.3 Target height, h , for 1% resolution at 100 DPI

h in Pixels	Menu Diameter						
Angle	4 cm	5 cm	6 cm	7 cm	8 cm	9 cm	10 cm
45°	0.61	0.77	0.92	1.07	1.22	1.38	1.53
60°	0.82	1.02	1.22	1.43	1.63	1.84	2.04
90°	1.22	1.53	1.84	2.14	2.45	2.76	3.06

The width of the target (w) is now defined (menu radius minus d_{21} or d_{101} as above). The height at the inner edge of the target has also been defined (one pixel). Next, the height (h) at the outer edge of the target is considered. Tables 3.2

and 3.3 show the target height, h , which occurs at the edge of the menu. Note that values of less than one pixel are in italic to indicate these values have zero width (the inner edge is at or beyond the edge of the menu.) Looking at the values in Table 3.3 that are greater than one pixel, 1% resolution can only be achieved with larger diameter menus (at least 7 cm), or menus with fewer items (4 cm with 4 menu items). Even with values that can be achieved, the target size is so small (h of two or three pixels) that the menu would probably not be usable.

Note that with an unbounded menu, both the width, w , and the height, h , of the target can potentially be much larger, limited only by the distance to the screen edge. However, since that distance is unknown and could potentially be less than the radius of the menu, if it is activated near a screen edge. But why is height important? Since only the overall menu item is visible, but the target for a given value (for example, the target for the value 5%) is not visible, there may be some circular seeking necessary (movement around the wedge to correct for an outward movement that results in the wrong value) to locate the desired value. During circular seeking, the target height represents width when applying Fitts' Law and it is therefore desirable to have a large h . Figure 3.6 shows an example of a user that has started toward a target with too high a value and had to switch to seeking in a circular direction to reach the desired target.

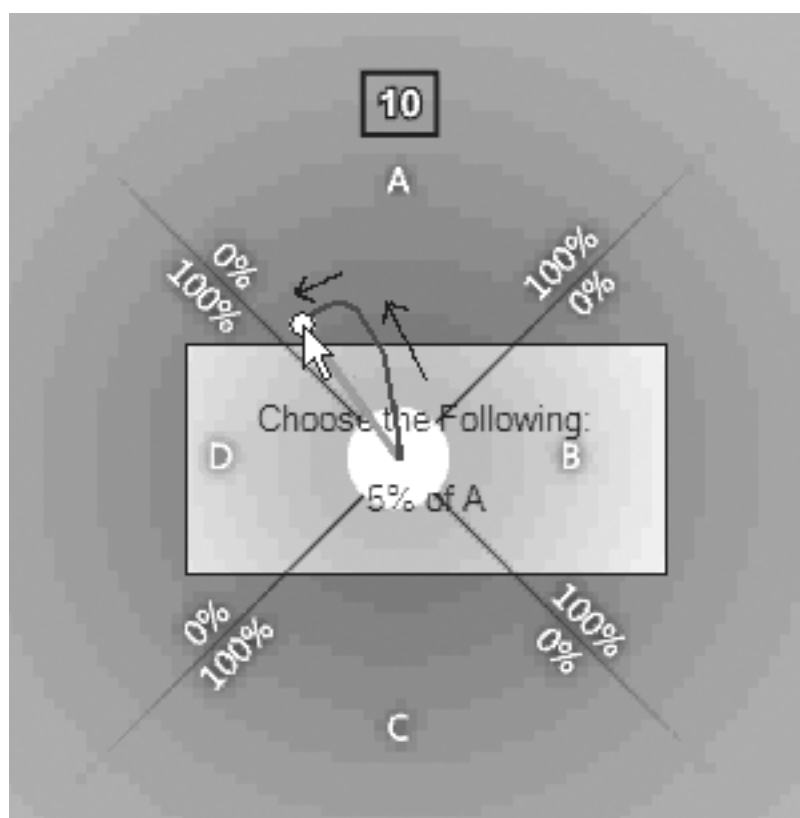


Figure 3.6 Example of Circular Seeking (from LoqSquare [46] Output)

More menu items allow for more selections, but as the number of items increases the target size will decrease and there will be an associated reduction in performance levels. Therefore only menus with four, six, or eight items were studied. Taking into consideration the limitations of target size at higher resolutions, only the 5% resolution control items were studied. Larger diameter menus may have a larger number of items, because of the larger target sizes they can accommodate. Larger items also allow more room for the text of the item providing better readability. However, this larger size menu must be balanced against larger hand movements and a greater possibility of the menu colliding

with the screen edge. Consideration was given to studying menus of various diameters. However, this would have increased the number of independent variables, and made the experiment take much longer than the timeframe this study would allow. In an attempt to find an optimal menu size, the experiment used an unbounded menu and the radius that was chosen by the participants was observed.

Even if the proposed menu is proven to increase productivity, it will not be adopted if it proves to be difficult to learn. To address the question of ease of learning, the experiment was structured to measure the rate of learning. The expectation was that the continuous control would follow the standard power learning curve [47].

Comparison to existing controls

How does this proposed control compare to previous controls? As can be seen from the related work in Chapter II, there are very few existing techniques that allow for continuous selection. Pie Menus, Marking Menus, Crossing Menus, Gesturing, Flicks, etc. are all menu/command based and do not allow for the selection of a value. Three methods were mentioned that were not command only methods: the Radial Scroll control, the FaST Slider, and Hopkins' continuous control methods.

The Radial Scroll Tool [41] , previously shown in Figure 2.12, allows a value to be changed. However, it does not allow a particular value to be selected

per se. The value, in this case the scroll position of a window, could only be increased or decreased. An absolute value could not be chosen directly. The proposed control allows a value to be chosen directly without the need to increment or decrement to reach the value.

The FaST Slider [40], previously shown in Figure 2.11, has several disadvantages compared to the proposed control. First, it takes two steps to get to the point of choosing a value: one step to display the control, and another to pin it to the desktop. It then takes two more steps before completion: one to select a value, and another to accept it. This complex series of selecting and steering takes time and imposes a heavy cognitive load on the user. The proposed control doesn't have these drawbacks.

The continuous control suggested by Hopkins [23] is similar to the control proposed here, but uses the distance from the center instead of the angle within a control. As was discussed previously in considering the design of the menu, the angle method is believed to be superior to the distance method.

3.2 Hypothesis

In this section, the basic questions raised previously are formalized into hypotheses. Those questions concerned the effect on performance of a visual mark; the direction of increasing value; the number of items contained in a menu; and the ease of learning the new control. These questions were answered empirically, and the process used to accomplish this is described below.

The following hypotheses were formed to answer the questions:

Question: Does a visual mark improve performance?

Hypothesis 1. A visual mark will yield a faster response time and a lower error rate than no visual mark.

This is based on the assumption that a visual mark will assist the user in understanding that the angle of selection is the controlling factor, and not the distance from the center.

Question: Which orientation of increasing value is best?

Hypothesis 2. A top-to-bottom orientation will yield a faster response time and a lower error rate than a clockwise orientation.

This is based on the assumption that for most, if not all languages, a page is read from top to bottom.

Question: How many continuous menu items can a menu contain?

Hypothesis 3. Collectively, users will tend toward a radius that is neither too close to the center, nor too far from the center. This will happen with a relatively small standard deviation.

This prediction is based on the fact that user's tend to click near the center of a target. [48] Since the menus in the experiment are unbounded, the concept of

“center” can be difficult. In this case the user will venture some distance into the menu to avoid the edge, since the edge is clearly not the center. The user will also avoid moving too far to avoid the increased selection time. Each individual will form their own concept of the most comfortable location and have a strong tendency toward that distance.

Hypothesis 4. Menus that subtend a larger angle will yield a faster response time and a lower error rate than menus that subtend a smaller angle.

A larger angle results in a target that is higher. Since the height of the control represents the width parameter in Fitts’ Law’s formulae during circular seeking, Fitts’ Law predicts a higher performance characterized by a faster response time and a lower error rate during this circular seeking.

Question: Is a continuous pie menu easy to learn?

Hypothesis 5. Selection time will quickly decrease then level off with use.

An initial decrease in selection time should take place while the users learn the interaction technique for this particular control. After the technique is mastered the performance should reach a more stable level. This follows a standard power learning curve.

3.3 Experiment

3.3.1 Variables

The following independent variables were controlled for this experiment:

Table 3.4 Independent Variables

Variable	Level	Fixed/Random	Description
Mark	Nominal	Fixed	Visible mark or none
Increasing Orientation	Nominal	Fixed	Clockwise or Top-to-Bottom
Number of items	Nominal	Fixed	4, 6, or 8 items
Menu Item	Nominal	Random	Which menu item
Value	Ratio	Random	Value to choose

The following dependent variables were measured:

Table 3.5 Dependent Variables

Variable	Level	Description
Time	Ratio	Time from activating the menu until a selection is made
Errors	Ratio	Number of incorrect selections (wrong value or wrong item selected) plus number of cancellations (menu dismissed before a selection takes place)
Selection Radius	Ratio	The distance in pixels from the center to the position where an item is chosen

Mouse trails of the participant's movements were collected. Although not part of the formal analysis, this information would be useful for examining the amount of circular seeking. It also acted as backup data in case the data from the experimental tool was corrupted.

The following demographics were collected, but were not used in the formal analysis performed:

Table 3.6 Demographics Collected

Variable	Level	Description
Gender	Nominal	Male or Female
Handedness	Nominal	Right or left handed
Pie Menu Usage	Nominal	Is familiar with pie menus or not

The following restrictions were placed on the participant to reduce variability in the data while reducing the number of factors studied. This limiting of the experiment reduced the number of participants needed for the study, which reduced costs.

Table 3.7 Restriction Applied

Variable	Level	Description	Restriction
Frequency of computer use	Ordinal	In the past month, how frequently was a computer used by participant	Must use a computer most every day
Age	Ratio	Age of the participant in years since birth.	Must be 18 years old or older

3.3.2 *Detailed Data Collection Plan*

3.3.2.1 *Participants*

Nine test participants, two female (participants 5 and 9) and seven male, were recruited from Wacom Technology Corp., the place where the principal investigator is employed. All participants were right handed. Four of the participants (2, 3, 7, and 9) had previously used a program that utilized radial menus.

3.3.2.2 *Equipment*

Testing was done on a Gateway® Model TA1 computer running Windows® Tablet PC Edition with the addition of an external mouse. The mouse used default settings including acceleration and speed. The display on the TA1 is a 14.0" wide LCD with a WXGA (1280 x 768) resolution and a dot pitch of 96 DPI. The application written to present the trials to the participant also recorded the dependent variables. The LogSquare tool from Mangold International [46] was also used to record trials.

3.3.2.3 *Task*

Before testing began, the participants were asked to fill out a consent form. The participants were then encouraged to place the computer and mouse in a position they considered comfortable. The participants were reminded of linear menu usage by bringing up the contextual menu for a file on the

Windows® desktop. The participants were then trained in the use of pie menus by selecting a few menu items in Second Life®. [45]

Testing consisted of presenting the participant with various combinations of the test menus described previously. The participants were asked to choose items as quickly and accurately as possible. Each trial began with an initial screen consisting of a large empty document with a box in the middle. Inside the box were the instructions for this trial. An example is shown in Figure 3.7 below. The participant then right clicked on the box to activate the menu.

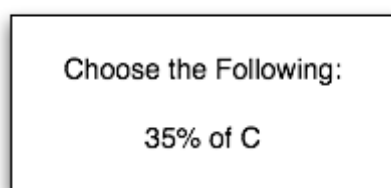


Figure 3.7 Sample test display

The menu items consisted of letters of the alphabet rather than a word to avoid any familiarity with menus in other applications. Other menu items were considered, such as animal names, but since there was to be a value associated with the items, words that are not normally described with values were considered inappropriate (what is 45% of an ape?). Conversely, items that might possibly have familiar values associated with them by the participants, such as volume or opacity, were also rejected. Therefore, letters of the alphabet were chosen to avoid positive or negative transfer for other programs the participant

might be familiar with. The menu layout remained static (had the same layout) for a series of trials. These restrictions helped to keep the task from becoming one of memorization or visual seeking. The layout of a 4-item menu with a visual mark and clockwise orientation as shown in Figure 3.8 was one of the layouts used in this experiment.

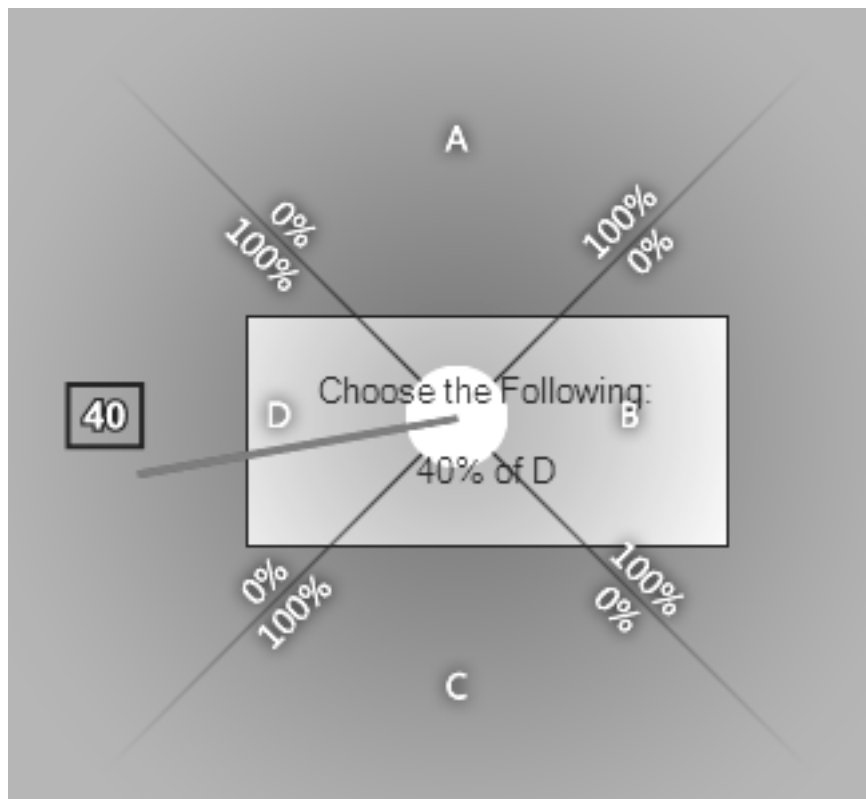


Figure 3.8 Experimental menu layout

Once the menu was activated, the participant then chose the indicated menu item and value. If the participant chose the wrong item, or a wrong value, the trial was recorded as an error. There was no feedback regarding right or

wrong choices, and there was no opportunity to make corrections. The test script used for the experiment is shown in Appendix A.

3.3.3 *Proof Criteria*

When comparing means, data analysis followed the standard practice of attempting to prove the null hypothesis. The null hypothesis was rejected when it was below a 5% confidence level (represented as $p < 0.05$), which is equivalent to proving the experimental hypothesis to a 95% confidence level. Significant results had a minimum power of 0.80. Had insufficient power been achieved additional participants would have been added up to a maximum of 18 participants. As would be expected, results which were not significant had lower power. Had post hoc analysis shown that a power of 0.80 could be achieved with additional participants, up to 18 participants would have been tested. When examining standard deviation, a standard deviation of less than one half of the mean was considered significant. When analyzing regressions, an R^2 of greater than 0.75 was considered significant.

3.3.4 *Experimental Design*

All participants performed all tasks, only the order of the tasks was changed, therefore this is a within-subjects experiment. To keep the experiment balanced, test participants were assigned to one of three groups on a rotating basis. The three groups were partitioned based on which test block they ran first

as can be seen in Table 3.8. The first test block consisted of menus with no visual feedback mark and a top-to-bottom orientation for numbering. The second test block consisted of menus with no visual feedback mark and a clockwise orientation for numbering. The third test block consisted of menus with a visual feedback mark and clockwise orientation numbering.

All participants completed the remaining blocks in random order, but balanced to provide the same total number of tests of each test condition. Note that the top-to-bottom orientation with visual feedback mark is not covered. An interaction between the visual mark and the direction of numbering was not expected, so this combination was dropped to reduce the size of the experiment.

Each test block consisted of three sets of tests, except the first, which consisted of four. The additional block was a training block and used to measure the learning rate. Each test set consisted of one combination of a number of menus items: four, six, and eight. The sets were presented in a random and balanced order. The very first set presented to the participants was repeated. This first set was a training set to allow time for the participants to become familiar with the menus. The first two sets were combined to measure learning. When measuring menu performance, the first set was excluded. The order of test sets for up to 18 participants is shown in Table 3.8.

Table 3.8 Testing Order

Participant #	First Test Block	Order of # of items	Second Test Block	Order of # of items	Third Test Block	Order of # of items
1	No mark, top-to-bottom	4, 4, 6, 8	No mark, clockwise	6, 4, 8	Mark, clockwise	8, 4, 6
13		4, 4, 8, 6		6, 8, 4		8, 6, 4
7		6, 6, 4, 8		8, 4, 6		4, 6, 8
14		6, 6, 8, 4	Mark, clockwise	8, 6, 4	No mark, clockwise	4, 8, 6
2		8, 8, 4, 6		4, 6, 8		6, 4, 8
8		8, 8, 6, 4		4, 8, 6		6, 8, 4
9	No mark, clockwise	4, 4, 6, 8	No mark, top-to-bottom	6, 4, 8	Mark, clockwise	8, 4, 6
15		4, 4, 8, 6		6, 8, 4		8, 6, 4
3		6, 6, 4, 8		8, 4, 6		4, 6, 8
10		6, 6, 8, 4	Mark, clockwise	8, 6, 4	No mark, top-to-bottom	4, 8, 6
16		8, 8, 4, 6		4, 6, 8		6, 4, 8
4		8, 8, 6, 4		4, 8, 6		6, 8, 4
11	Mark, clockwise	4, 4, 6, 8	No mark, top-to-bottom	6, 4, 8	No mark, clockwise	8, 4, 6
5		4, 4, 8, 6		6, 8, 4		8, 6, 4
17		6, 6, 4, 8		8, 4, 6		4, 6, 8
6		6, 6, 8, 4	No mark, clockwise	8, 6, 4	No mark, top-to-bottom	4, 8, 6
18		8, 8, 4, 6		4, 6, 8		6, 4, 8
12		8, 8, 6, 4		4, 8, 6		6, 8, 4

Each set required the participant to make 24 item choices from a menu. Each item in each of the menus was chosen an equal number of times in a permuted order. For a four-item menu, each item was chosen six times; for a six-item menu, each item was chosen four times; and for an eight-item menu, each item was chosen three times. For each item selected, the participant was

instructed to choose a particular value from 0 to 100 in increments of 5. Which value the participant had to choose was randomly determined.

This design resulted in a total of 240 discrete tests per participant. Assuming an average of 3 seconds per test, this would require 12 minutes of testing per participant. The overall test session actually took longer because it included the completion of the consent form, instructions, and some short breaks between test sets. The goal was to keep the test session to less than 30 minutes per participant, which was achieved.

3.3.5 *Critique of the Design*

There are many factors that could have impacted the experiment. Certainly the visual appearance of the menus (label font, font size, boundaries between menu items, menu item background, etc.) and visual feedback (location, font, font size, highlighting of the current item, etc.) could have had an impact. The input device could have had an impact. Neither the use of the pen nor the track-pad that were supplied with the test computer was studied. Using a pen for input has been shown to be faster in some experiments [49], but this may not hold true for pie menus since the hand obscures a large portion of the menu.

There are factors that were controlled through randomization that could instead have been included as variables. For example, location of an item within the menu is known to have an impact on performance [11]. In this case, all items were chosen an equal number of times instead of making each item's position an

independent variable. The reason for controlling, rather than including these variables in the experiment, was to keep the size of the experiment manageable. Also, the factors were not truly randomized due to the balancing and blocking of the experiment. These restrictions on randomization could have had an effect on the results. The lack of randomization of the selection of participants could have been an issue since they were all employees of one company from one location. However, this had the advantage of reducing the variability in the experiment by using a single user class. If the control was found to be unusable with one user class, expanding to a larger number of user classes becomes unnecessary.

CHAPTER IV

ANALYSIS AND RESULTS

4.1 General Comments

Outliers were not discarded in any of the analyses. In one case, there was evidence that selection radius outliers were clearly valid data. Since there was no obvious reason to discard outliers in the other analyses, all outliers were retained. Tests where the participant made an error, such as an incorrect selection, were excluded for all analyses except when analyzing errors.

Selection time data was not normally distributed. Upon experimentation, however, it was found that the natural log of the time data was normally distributed. This is not surprising, given that the formula for Fitts' law includes a natural log. Since the value of interest is time, and not log time, time was used in the analyses. Furthermore, the analysis method chosen is fairly robust in the presence of non-normality

All probability measures were computed using SPSS version 16 [50]. The specific analysis performed was a general linear model repeated measures. This is a variation of the Analysis Of Variance technique. This analysis is used to compare sample means in the presence of repeated measurements. Results

indicate the probability that the means are the same (which represents the null hypothesis, since the experiment is trying to prove a difference in means) and are reported in the form $p < 0.12$. This would represent that there is a 12% chance that the means are the same. To reduce the degrees of freedom of the repeats, which was necessary to match the number of participants to the available degrees of freedom, the mean of each test set was used. If this were not done, over 30 participants would have been necessary due to the 24 repeats for each test.

SPSS does not lend itself well to analysis of an unbalanced experiment. To satisfy the requirement for a balanced experiment, two separate analyses were performed. One analysis used the no-mark top-to-bottom data and the no-mark clockwise data. This provided results for the direction and number of items. The other analysis used the no-mark clockwise and mark clockwise data. This provided results for the mark and number of items. Since both analyses provide results for the number of items, results were shown for both analyses when examining the effect of number of items.

4.2 Mark Effects

There was an expectation that a visual mark would act as a cue to the operation of the menu, and that this cue would improve the user's performance. The results of the analysis however showed that the presence or absence of the mark actually had no effect on performance time ($p < 0.362$). The inclusion of 0% and 100% visual range indicators on the item borders may have been sufficient to provide the necessary cue. Figure 4.1 shows the times for various numbers of items for the mark and no mark conditions. As can be seen in Figure 4.1, there was a noticeable reduction in variability of time with the mark, but no difference in the means for the same number of menu items.

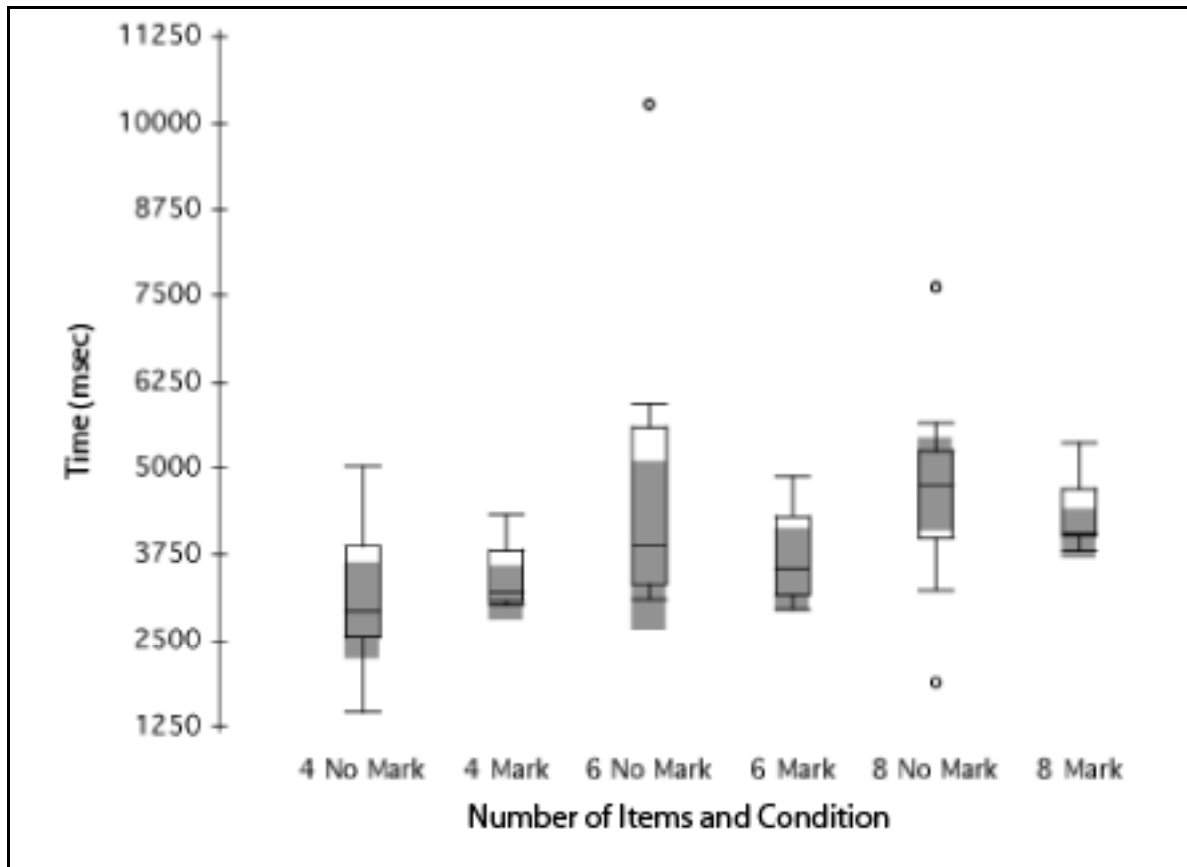


Figure 4.1 Times for No Mark and Mark Conditions²

² Note: In the box plots, the shaded areas indicate 95% confidence intervals for the mean. The line through the box is the mean. The extents of the box are 25% and 75% (mean of the bottom half of the data and mean of the top half of the data). The whiskers represent the extent of the data. Outliers are circles. Extreme outliers are asterisks.

The visual mark also had no effect on the error rate. As can be seen in Figure 4.2, the numbers of errors were almost identical with and without the mark ($p < 0.382$).

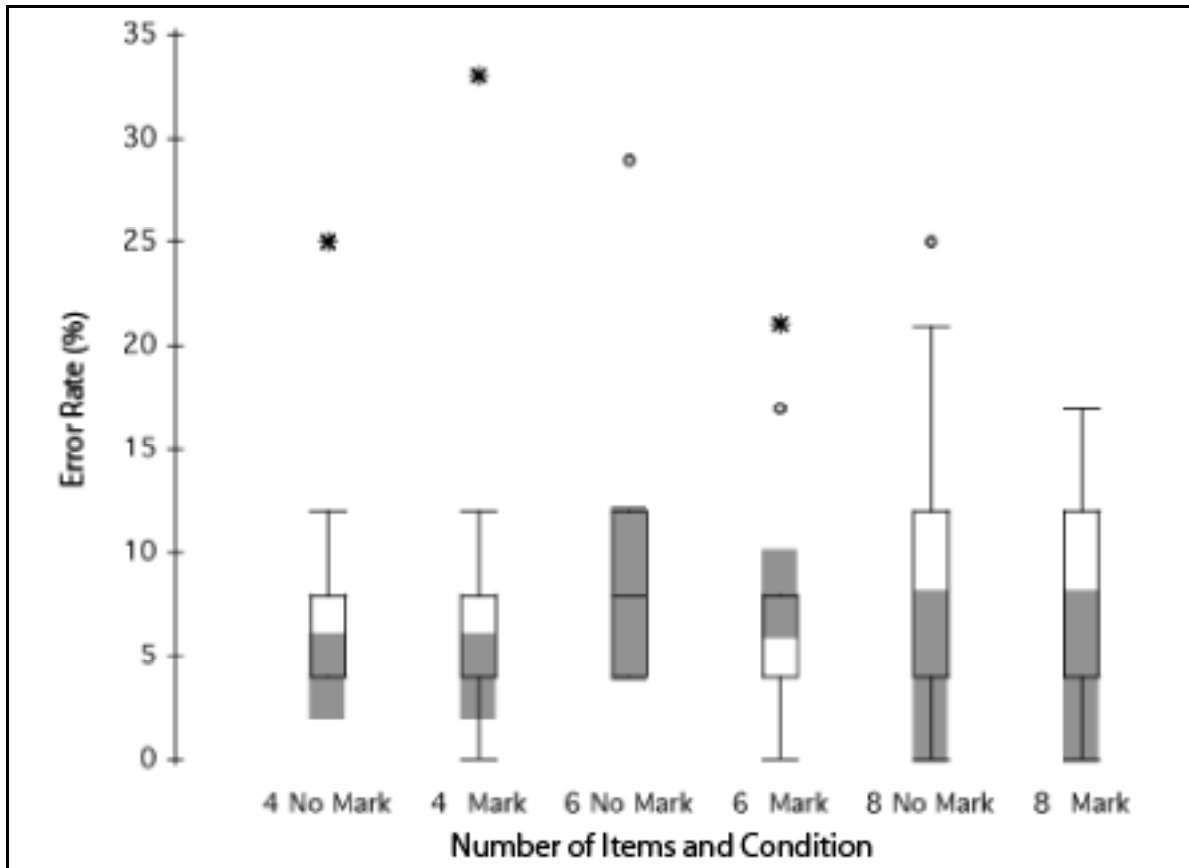


Figure 4.2 Percent Errors for No Mark and Mark Conditions

There was no hypothesis regarding selection radius and the mark, but there was some evidence that with the inclusion of the mark, the selection radius increased with near borderline significance ($p < 0.069$ with a power of 0.455).

Perhaps there was something visually appealing about a longer mark. This aspect might be worth future investigation.

Hypothesis 1 stated: “A visual mark will yield a faster response time and a lower error rate than no visual mark.” Since there were no significant differences in time or errors when the mark was included, the user’s performance was not impacted. Therefore the null hypothesis is accepted and the experimental Hypothesis 1 is rejected.

4.3 Orientation

Analysis of the orientation of the values within a wedge presented a bit of a challenge. Should all menu items be considered, or only those items where the orientation makes a difference? For example, in a menu with four items, only one of the four items, item “D” in the experiment, had a different orientation between the two tests. Selecting all items an equal number of times was done to reduce the confounding influence of direction [51]. Does restricting the analysis to only those items that are different allow the direction to confound the results?

Upon examining the data, the lack of a directional effect rendered the above question moot. The different orientations had no effect on time. This was true both when considering all items ($p < 0.724$) and when looking only at items where orientation made a difference ($p < 0.124$). Figure 4.3 shows the times for top-to-bottom and clockwise orientations for various numbers of items where all items were included.

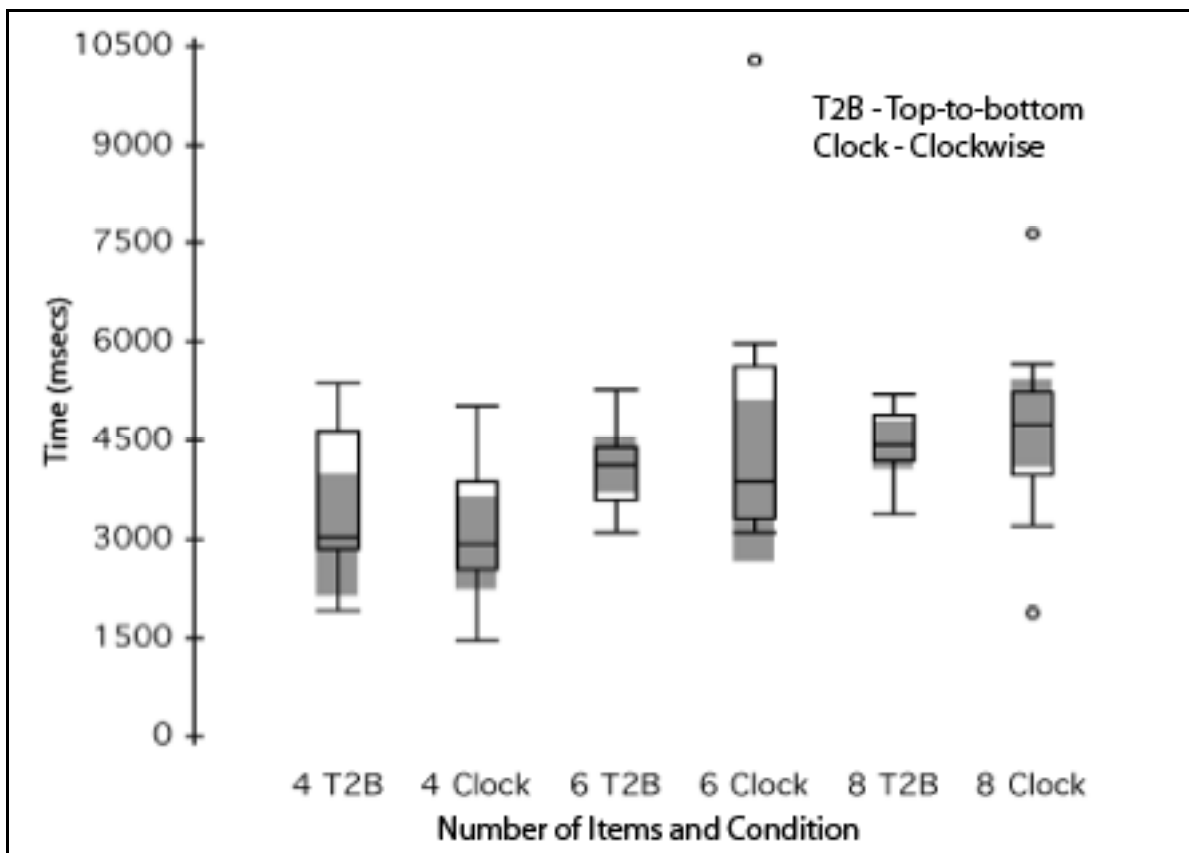


Figure 4.3 Times for Top-to-bottom and Clockwise for All Items

Figure 4.4 shows the times for top-to-bottom and clockwise orientations for various numbers of items. Only items where direction made a difference are shown in this Figure.

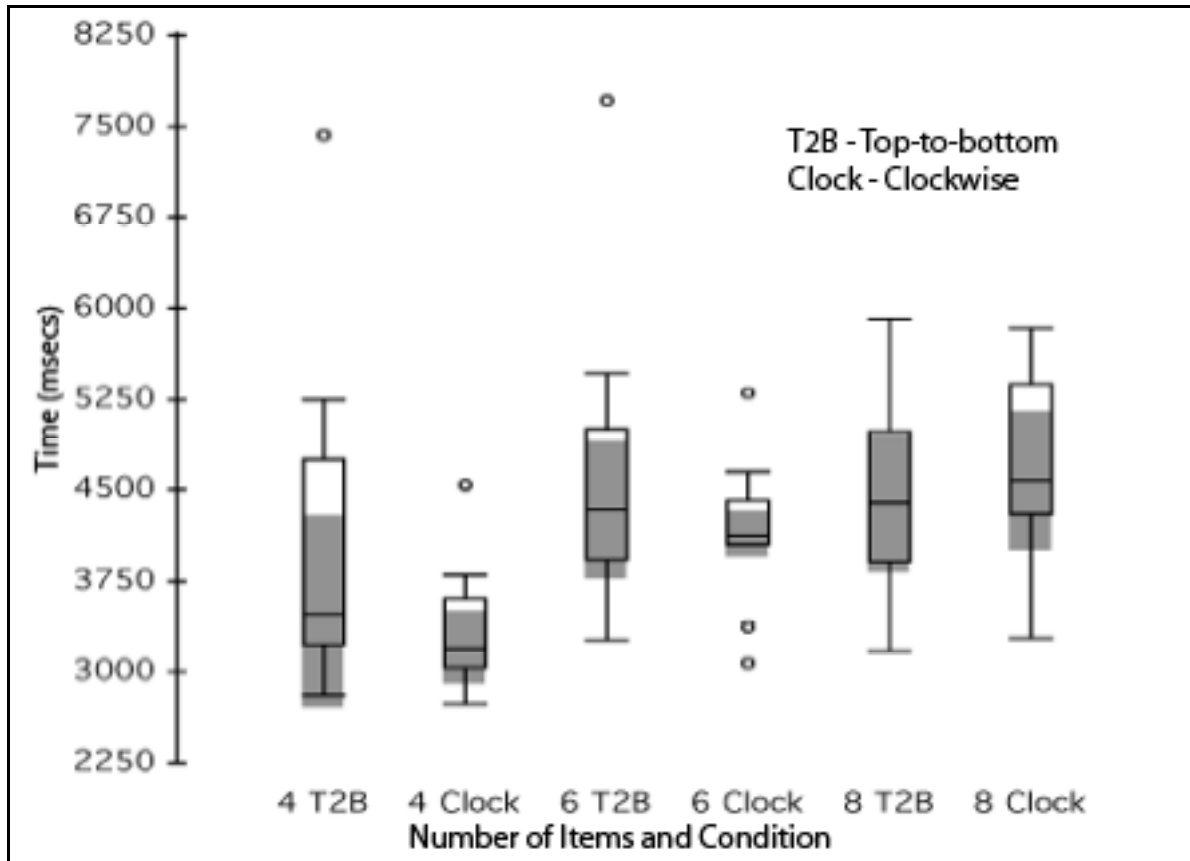


Figure 4.4 Times for Top-to-bottom and Clockwise for Directional Items

The clockwise orientation showed slightly fewer errors, but the numbers of errors were not significantly different between the top-to-bottom and clockwise orientation conditions ($p < 0.076$). Figure 4.5 shows the error rates for the top-to-bottom and clockwise orientations for various numbers of items where all items are included.

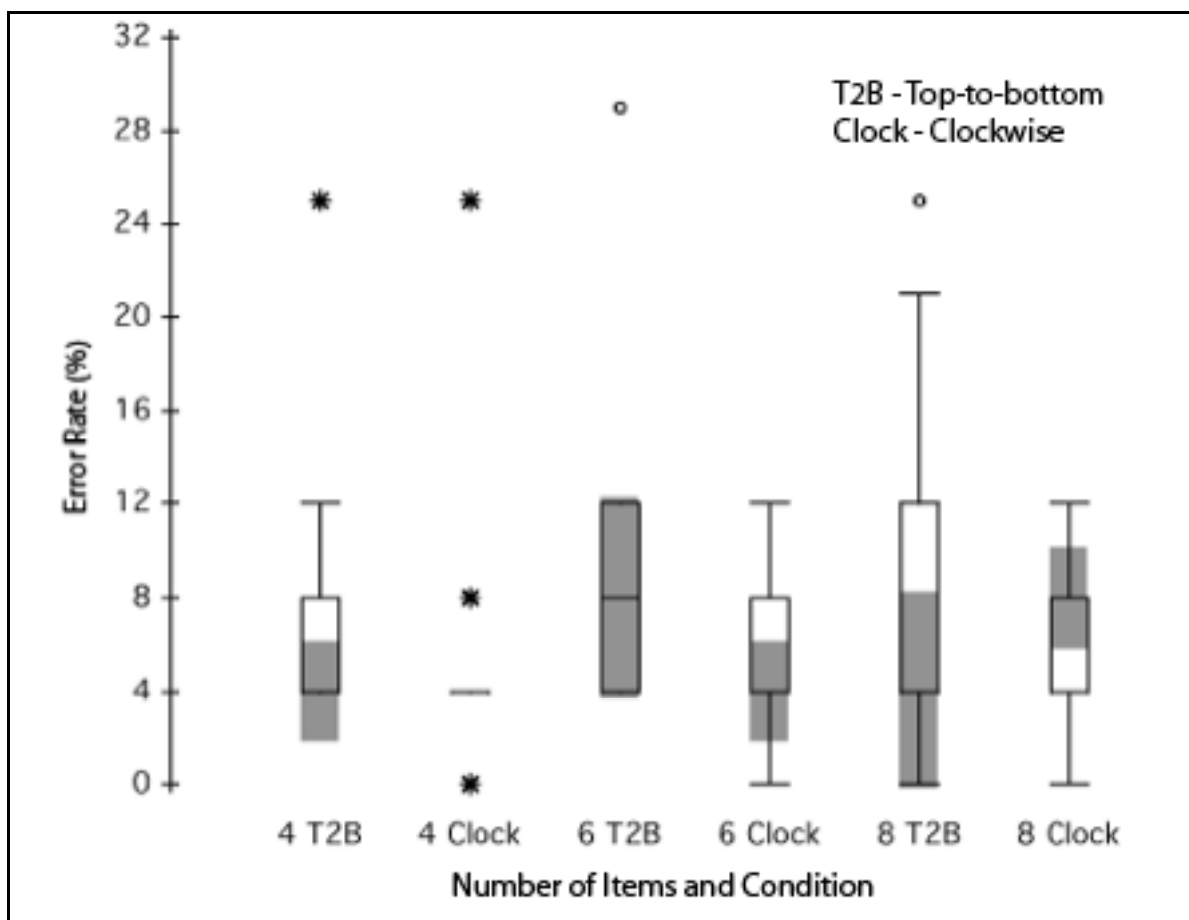


Figure 4.5 Percent Errors for Top-to-bottom and Clockwise for All Items

Hypothesis 2 stated: “A top-to-bottom orientation will yield a faster response time and a lower error rate than a clockwise orientation.” Since there

were no significant differences in time or errors for the two orientations, the null hypothesis is accepted and the experimental Hypothesis 2 is rejected.

There was no hypothesis regarding selection radius and orientation, but there was some evidence that with the clockwise orientation, the selection radius decreased. It did not rise to a level of significance ($p < 0.109$ when all items were considered) but those results may be inconclusive due to insufficient power (power < 0.355).

4.4 Selection Radius

Table 4.1 shows the mean selection radius across various test conditions. These values were computed without discarding any outliers. These relatively small radii, with standard deviations that are less than half of the associated selection radius, supports Hypothesis 3 which stated: “Collectively, users will tend toward a radius that is neither too close to the center, nor too far from the center. This will happen with a relatively small standard deviation.” Therefore, experimental Hypothesis 3 is accepted.

Table 4.1 Mean Selection Radii for Various Conditions.

Condition	Mean	Std. Deviation
Top-to-bottom, No Mark	52.59	25.85
Clockwise, No Mark	48.57	22.31
Clockwise, Mark	62.07	35.82
All	54.40	29.10

Although the results for selection radius were as predicted, the selection radius data merits further analysis. The data was skewed toward higher values, and had values that could be considered outliers (beyond two sigma) on the high side. On the high side, there were 116 cases beyond 2 sigma (112 pixels radius) and 36 cases were beyond 3 sigma (141 pixels). However, all but four of the values beyond three sigma were from the same participant, so these values are more indicative of participant preference and should therefore be considered valid data and not outliers. As can be seen from the mean values in Table 4.2, the selection radius was highly participant dependent. The lowest mean for a participant was 31.29 pixels and the highest was over 3 times higher at 97.73 pixels. The variability by participant is best seen in Figure 4.6, which shows the average selection radius for each of the nine test sets by participant. This preference for various radii by the participants should be considered during menu design.

It should be noted at this point that it is not clear whether the number of pixels or the physical distance is the important factor. The participants may have been choosing a distance based on the visual layout of the control, which would indicate the number of pixels is the important factor. On the other hand, it may have been the amount of physical distance that the participants tended toward.

Table 4.2 Selection Radius by Participant

Participant	Mean (pixels)	Std. Deviation (pixels)	Mean (cm)	Std. Deviation (cm)
1	80.63	25.637	2.13	0.678
2	44.05	11.527	1.17	0.305
3	97.73	38.354	2.59	1.015
4	77.63	18.376	2.05	0.486
5	43.09	12.077	1.14	0.320
6	37.61	12.337	1.00	0.326
7	31.29	7.544	0.83	0.200
8	44.05	11.298	1.17	0.299
9	39.28	13.006	1.04	0.344
Total	54.39	29.092	1.44	0.770

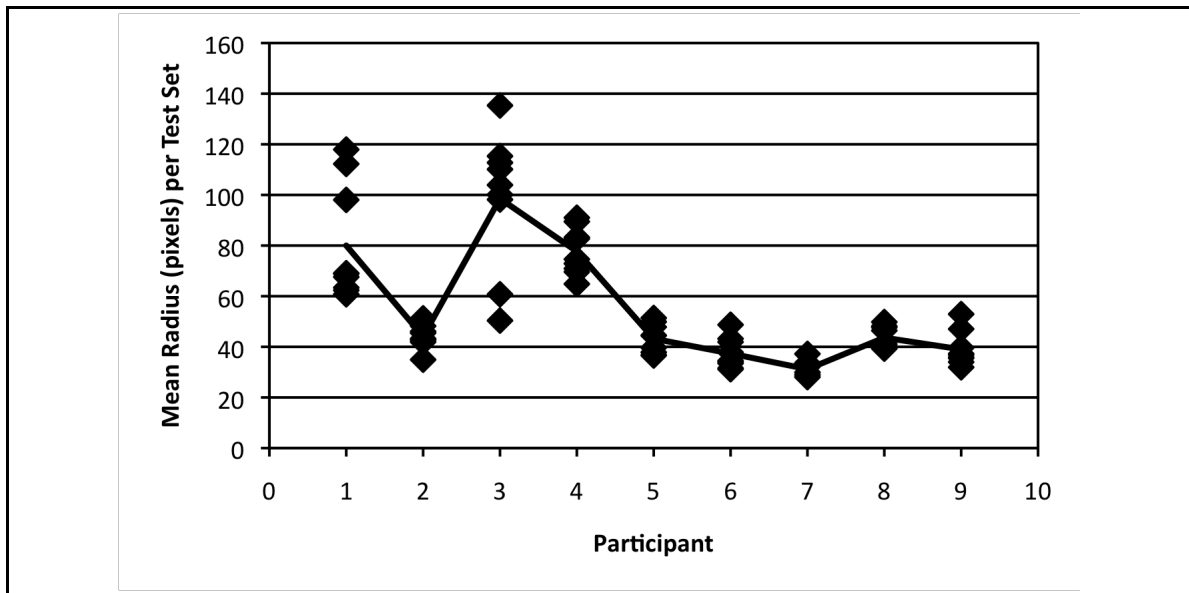


Figure 4.6 Mean Selection Radius per Test Set for each participant

4.5 Number of Items

One of the questions to be answered was “How many continuous menu items can a menu contain?” The conclusion reached by looking at the performance data for the number of items is that fewer items in the menu resulted in significantly lower time ($p < 0.007$ and $p < 0.009$). This can be seen in Figure 4.7 where the time for four items is clearly less than for eight items for similar conditions, with the time for six items falling between the two.

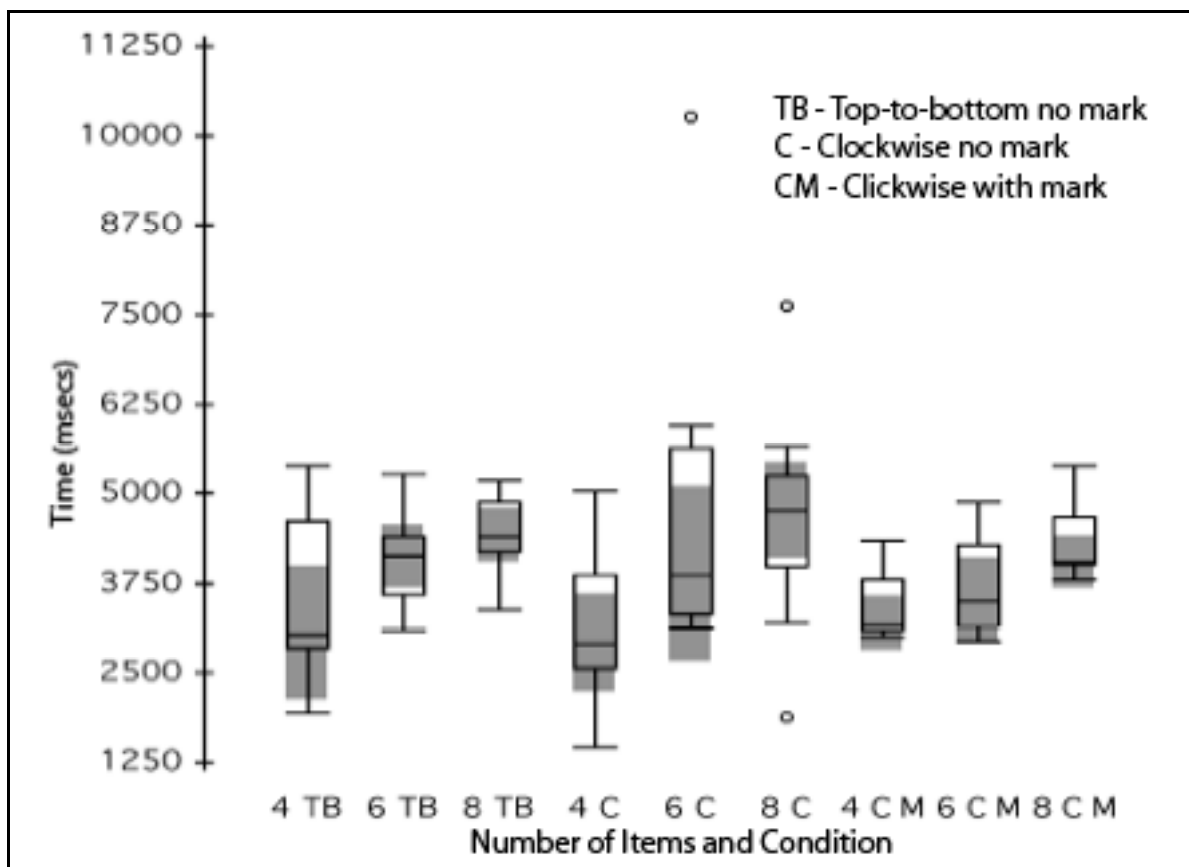


Figure 4.7 Selection Time for Number of Items Under Various Test Conditions

The number of items in the menu had no noticeable effect on the error rate ($p < 0.991$ and $p < 0.875$). Looking at Figure 4.8 it can be seen that the error rate for four items and eight items were almost identical. Although not significant, for two of the test conditions, the error rates for six items was worse than for either four or eight items. This might be explained by the orientation of the items. In both four and eight item menus, the items were located on compass points. This is supported by Kurtenbach's work on Marking Menus where the results indicated "... laying out items at the points of a compass or hour positions of clock can be used to counteract the increased difficulty of having many items in a menu". [34]

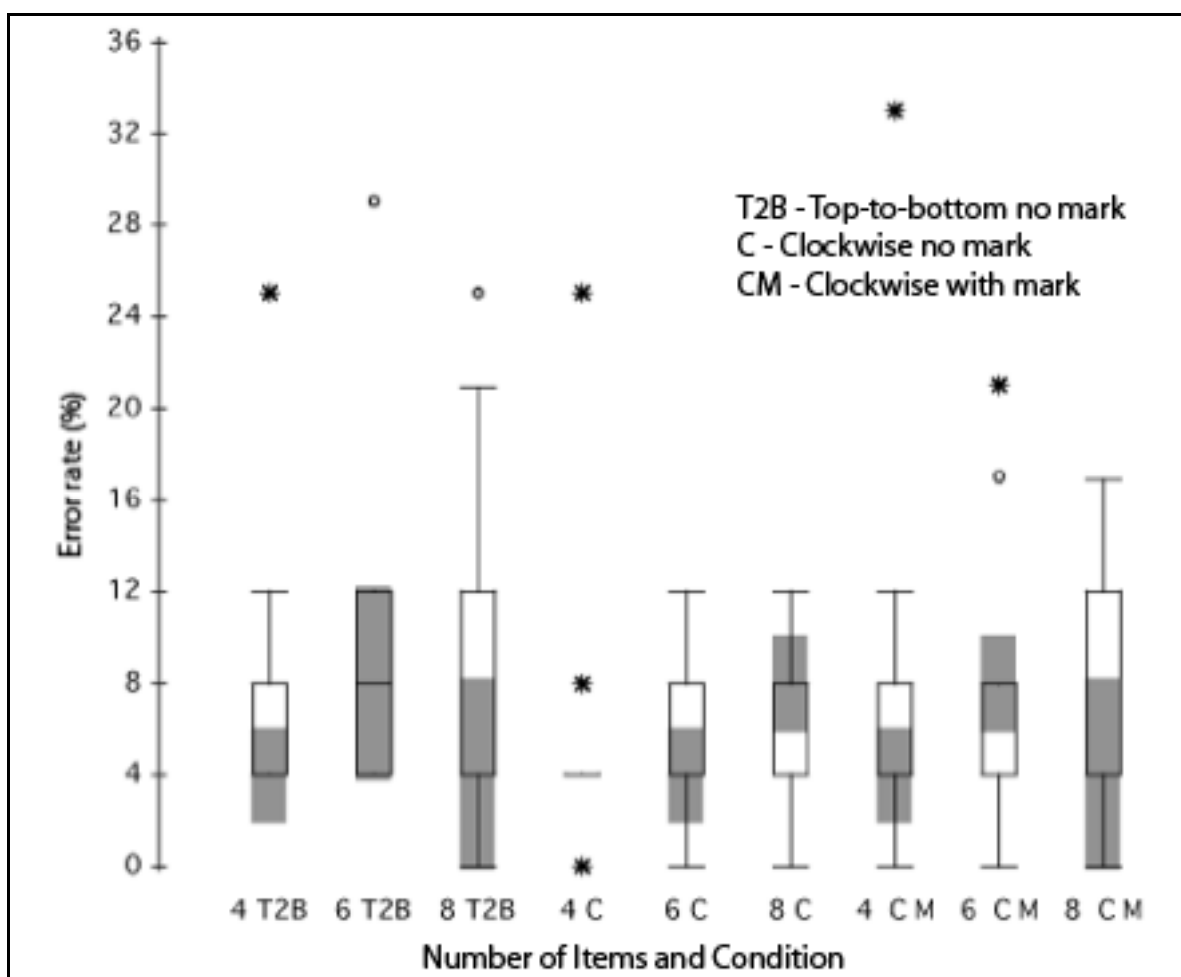


Figure 4.8 Percentage Errors for Various Test Conditions

As might be expected, errors were very participant dependent. Figure 4.9 shows the percent of incorrect choices made by each participant for each of the test sets. Figure 4.10 shows the average time required for each test of the nine test sets completed by each participant. It can be seen that the fastest participant was also one of the least error prone, whereas the slowest participant did not complete any test sets without error. This should not be construed as a deficiency of the control, rather as participant variability.

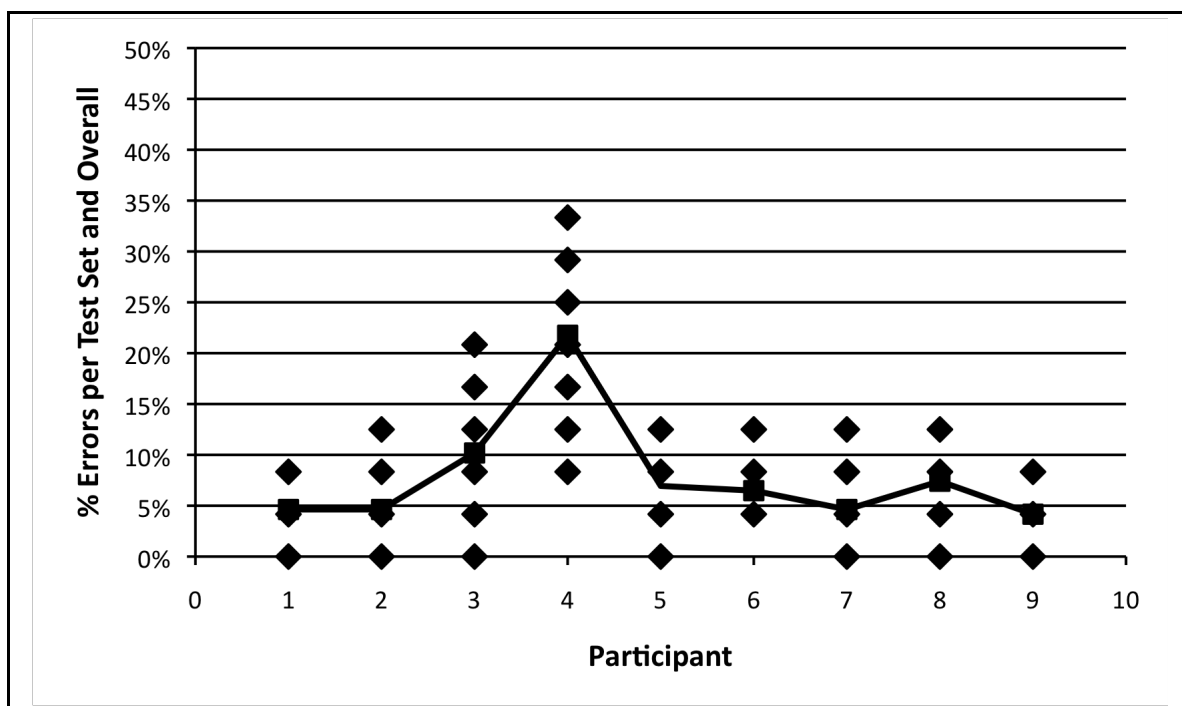


Figure 4.9 Percent Incorrect Selections per Test Set for each Participant

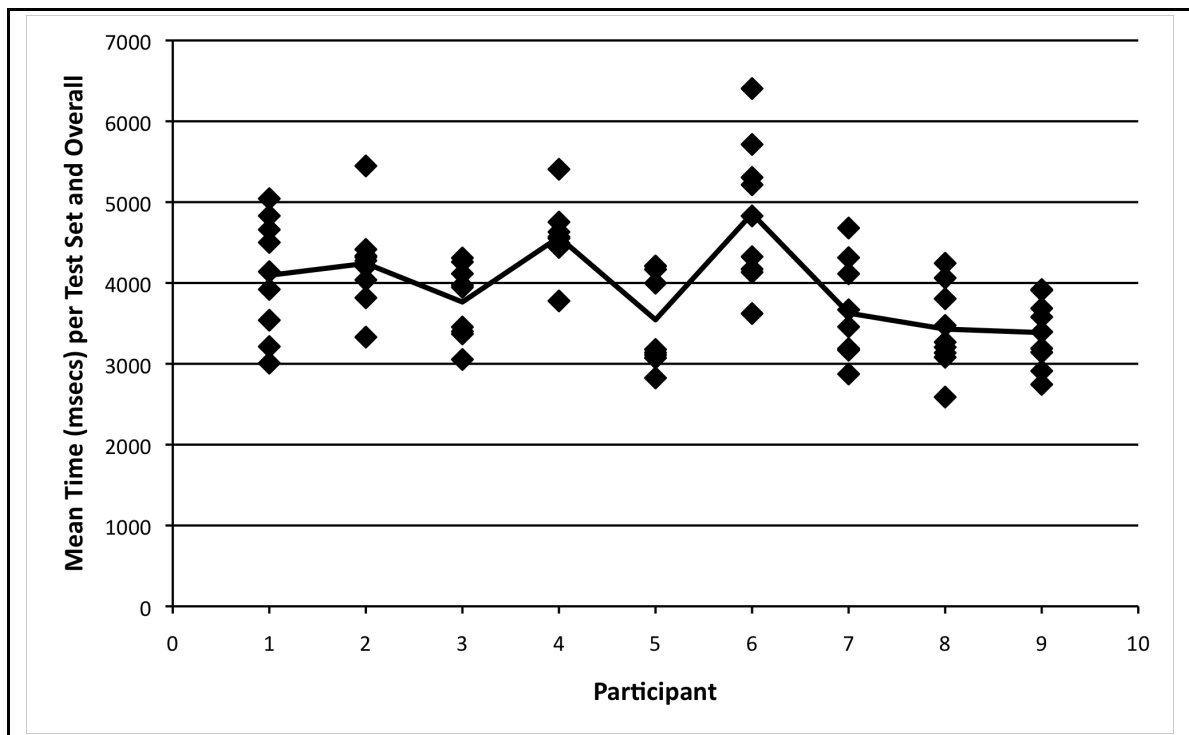


Figure 4.10 Mean Time per Test Set for each Participant

The data was examined for a correlation between errors and time. A negative correlation would be expected, as more careful participants are usually less error prone, but slower. Figure 4.11 shows a linear regression of error rate and time. The very low R^2 of 0.05 indicates that there was no correlation between errors and time for the given data. The low number of error samples, and the high variability between participants, may be confounding the relationship.

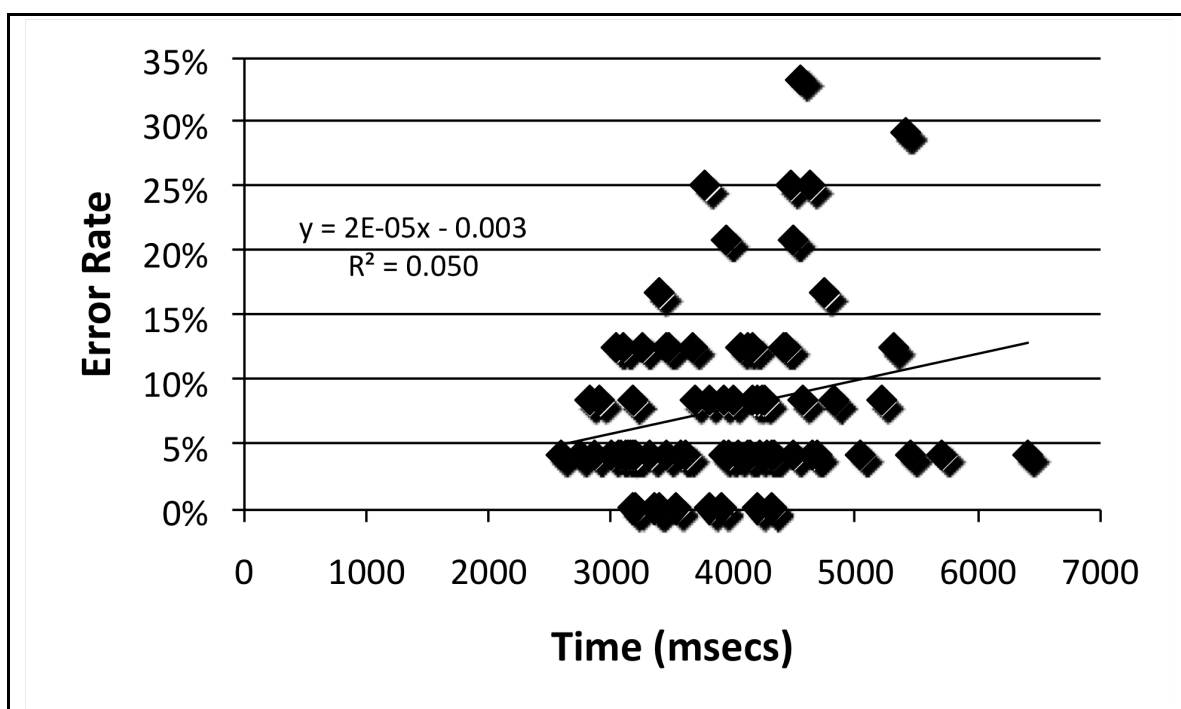


Figure 4.11 Relationship Between Time and Errors

Hypothesis 4 stated: “Menus that subtend a larger angle will yield a faster response time and a lower error rate than menus that subtend a smaller angle.” A menu with fewer items resulted in items that subtended a larger angle. Although the error rate was constant across the various numbers of items, the time was faster with a smaller number of items resulting in overall better performance with fewer items. Therefore, experimental Hypothesis 4 is accepted.

4.6 Learning

Participants began the experiment with no previous training. As noted above, the very first set of tests presented to a participant was repeated. This resulted in 48 repeats of the same test at the beginning of the experiment. To

examine the learning that took place, these 48 tests were broken into 8 blocks of 6 tests each. The mean time for each block was computed across all participants, which resulted in 54 tests per block. A power curve was then fit to the means. The R^2 of the fit is 0.8373, which demonstrates that learning closely followed the power learning curve. As can be seen in the graph in Figure 4.12, most of the learning took place in the first two blocks, or within the first twelve selections. This supports Hypothesis 5 that selection time will quickly decline then level off.

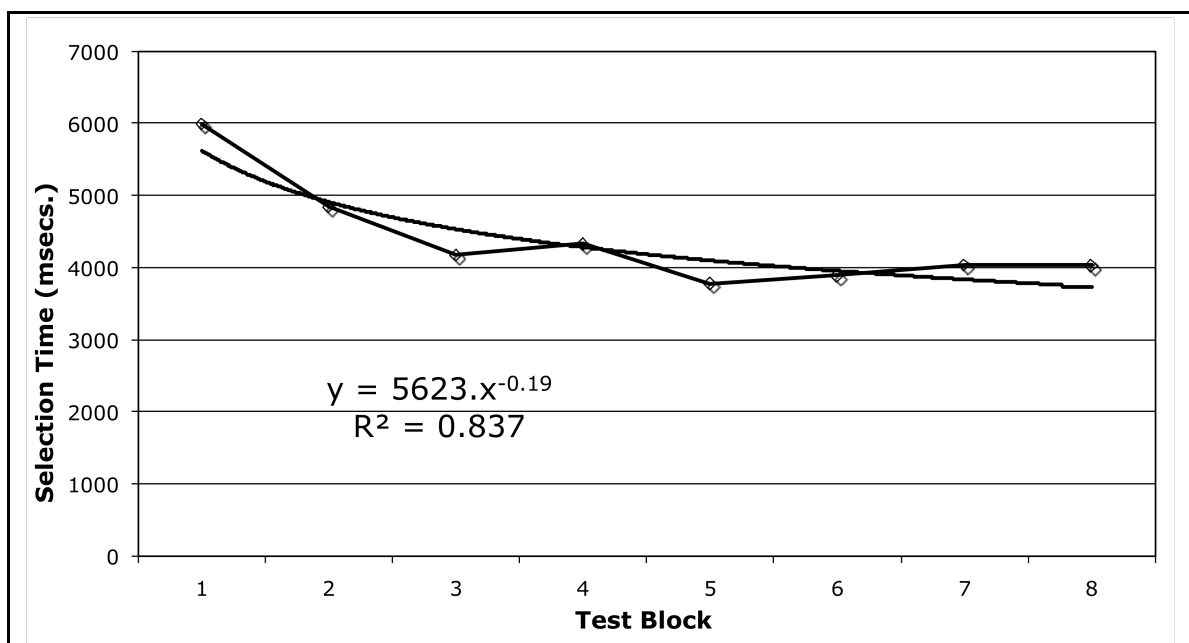


Figure 4.12 Mean Selection Time for Successive Blocks of Tests

CHAPTER V

CONCLUSIONS, SUMMARY, AND FUTURE WORK

5.1 Conclusions

In this section the results are considered, and how those results relate to the design and implementation of a continuous radial menu.

Hypothesis 1. The visual mark had no impact on performance, but the presence of the mark did encourage an increase in selection radius. Since angular resolution increases with a larger radius, this is a positive aspect of the mark. There was no attempt to investigate subjective measures of the participant's attitudes toward the menu. However, during tests done while piloting the experiment, one participant commented that they thought they were faster with the mark. Although this was not the actual case for this participant, or for participants in general, and was an unsolicited comment by a single participant, the perception of increased speed may be another positive aspect of the mark.

Design Implications. Given the lack of impact on performance, the results give no guidance as to whether the mark should be included or excluded. The decision whether or not to include the mark is therefore largely an aesthetic one.

The anecdotal evidence of perceived improvement, and the possibility of users choosing a larger radius during use, leans in favor of including the mark. There is no evidence that the mark provides the desired angular cue, but the 0% and 100% labels may have had this effect. Without further experimentation, the visual layout selected should be maintained in case it was providing usage guidance.

Hypothesis 2. The orientation of increasing value had no impact on performance. It appears that the strong top-to-bottom preference seen in written language is balanced by an equally strong preference for a clockwise orientation seen in clocks, mechanical dials, and meters. Although not significant, there were approximately two percent fewer errors with the clockwise orientation.

Design Implications. As with the mark, the decision on orientation becomes largely an aesthetic one. Unless more experimentation is done, a choice must be made to move forward. Since there were slightly fewer errors with clockwise, and a clockwise matches a knob, clock, or meter paradigm, the clockwise orientation is slightly preferable to top-to-bottom.

Hypothesis 3. As predicted, participants tended toward a favored selection radius. This favored radius was very participant dependent. The range of values obtained is useful in setting design guidelines for menus.

Design Implications. For a five percent control, dropping below the d_{21} distance makes some values impossible to obtain. To keep users that tend to favor a small radius from dropping below the d_{21} distance, the center of the control should have a diameter of d_{21} (substitute the proper d_{xx} as appropriate for different resolution controls). To accommodate users who favored a large selection radius, the minimum radius for a bounded menu should be at least 100 pixels. This translates to approximately 2.5 cm on the system used for testing. This is also an indication that this control is not well suited to very small displays. An unbounded menu allows for more flexibility and accommodates the preference of users that tend to favor large radii, which were at times as large as 120 pixels or more.

Hypothesis 4. More menu items lead to smaller items and longer selection times. Although the selection time for smaller items was statistically significantly smaller, it was not drastically smaller (“drastic” being a clear and noticeable difference impacting usability). Also, the error rate for the smaller items was not significantly worse. It would be interesting to experiment with a higher number of items see if there is a break in the curve toward drastically worse performance at some larger number of items.

Design Implications. Performance with four, six, or eight items was such that a menu design with any of these designs would be quite usable. The goal

was to see how many items could be used, since there is often a need to provide as many choices as possible. Given the results, the menu with eight items is optimum within the range of values studied. A 10 or 12 item menu also might be usable, but there is no data to support these higher numbers.

Hypothesis 5. Given that there was no prior training, the menu proved to be easily discoverable and learnable. The coder collecting the data noted that participants tended to have an “ah ha” moment where the control was suddenly understood. This tended to happen within the first half dozen selections. Table 5.1 shows the selection times for the very first six selections for all nine participants. As can be seen by the “*” error tags in Table 5.1, only two of the nine participants had errors in the first six selections. Most participants took longer for their very first selection as they explored how the control worked. There is insufficient data to examine the effects of the mark or direction on learning. However, even the limited data has no clear trend (for example, participants five and six had a mark for their first trial, seven and eight did not, yet five and seven did well, and six and eight did not.)

Table 5.1 Selection Times (msecs.) for Initial Selections

Selection	Participant								
	1	2	3	4	5	6	7	8	9
1st	*	22110	6063	10953	7000	25719	6812	18437	9969
2nd	*	5313	7125	3641	8469	*	7266	3484	4157
3rd	*	3625	4328	2782	5000	3344	4234	3484	3859
4th	18359	4844	3422	2484	5219	1859	3547	4672	3297
5th	4313	2407	5547	2078	6890	7938	2985	3719	3234
6th	3110	4297	3781	4360	2890	4828	4032	5250	3000

Design Implications. The data is more of a validation of usability than a guidance of design parameters. Since most participants had mastered the control after a single click, the control appears to be quite usable.

Other Design Implications. The implementation of the control used in the experiment has a numeric indicator that should be retained. Whenever the screen cursor was in the area of an item, a numeric readout appeared. The fact that this indicator appeared only in the item the user was currently in is a hint as to which item will be selected. This readout showed the current value based on the screen cursor location. Knowing the value was critical to selecting a value, and is therefore an integral and important part of the control.

5.2 Summary

With large displays, especially large interactive pen displays, the large hand movements needed to reach menus and palettes located at the screen edge can result in low productivity and fatigue. Solutions exist to bring the interface to

the user. One such solution is using gesturing, where a gesture can be performed at any location, not just at the screen edge. Another existing solution is using pie menus, where a menu containing a list of selections is presented at the current location. However, these and other existing solutions are menu or command centric. Few solutions exist that allow the user to select a continuous value, and those that do have distinct disadvantages.

This research has resulted in a design for a new continuous control for inclusion in radial or pie menus. In the process of designing the new control, several questions were raised. Those questions are: Does using a visual mark improve performance? How many items can a menu contain? Is the new control easy to learn? Which direction should the values increase? A series of hypotheses were developed to formalize these questions. In order to verify the hypotheses an experiment was designed. The results of the experiments verified the usability of such a control and of some of the design parameters of such a control. The results favor the following recommendations:

- A numeric indicator showing the current value must be provided.
- A visual mark showing the current angle within the control should be provided.
- The numbering should be such that values increase in a clockwise direction.
- Minimum and maximum numeric indicators should be provided on item boundaries.

- The center of the control should be dictated by the resolution of the control and size of the items. The diameter should be such that the user is assured of the ability to select all values
- The outside should be unbounded in order to accommodate users that prefer a large selection radius.
- If there are a large number of choices to be made, a menu with eight items is usable. More items may be possible, but there is no data to support more items.

A menu that incorporates the above design considerations was shown to be easy to learn. Such a menu provides a means of selecting more than a simple command or single attribute, and also allows for a choice from a range of values. Combined with existing radial menus, even more of the user interface can be brought to the user, thus increasing productivity and providing a richer user experience.

5.3 Future Work

As might be expected with formal research, resource limitations leave many questions unanswered, plus more questions are raised by the research. Some of those questions are listed here:

- How many items can a menu contain before it becomes unusable? With up to eight items there was no inflection point where usability became drastically

worse. It would be interesting to try ten or twelve items to examine the usability of even denser menus.

- How well does the control work when it is not at the top level of the hierarchy? (For instance, when a parent menu is selected to show a submenu containing this control.)
- Many interactive displays are stylus based, or at least use a stylus like device, such as markers for electronic whiteboards. What are the performance and usability difference between a mouse and a stylus, or even other input devices such as track-pads, track-sticks, or trackballs?
- How the control works in “real life” is an unknown. Would the performance change if the items have meaning (such as adjusting volume, width of drawn lines, etc.)?
- Subjective measures were not taken. How acceptable is the new menu on a subjective basis? Will users say they like it, or more importantly, will users actually use it, if given a choice?
- What about different user classes? Exploring users of various age groups and levels of computer experience would be valuable in verifying that the control is usable in the general population.
- The current control uses a right click and release to show the menu, and a left click release to make a choice. Would there be a performance enhancement if

a right click showed the menu and a right release made the selection? Would this technique be discoverable?

Further experimentation to address any of these questions would be beneficial in further refining the usability of this control.

REFERENCES

- [1] P. M. Fitts, "The Information Capacity of the Human Motor System in Controlling the Amplitude of Movement," *Journal of Applied Psychology*, vol. 47, pp. 381-391, 1954.
- [2] C. E. Shannon, "A Mathematical Theory of Communication," *The Bell System Technical Journal*, vol. 27, pp. 379-423, 623-656, July & October 1948.
- [3] I. S. MacKenzie, "Movement Time Prediction in Human-Computer Interfaces," in *Proceedings of the conference on Graphics Interface '92*, Vancouver, British Columbia, Canada, 1992, pp. 140-150.
- [4] J. Accot and S. Zhai, "Beyond Fitts' Law: Models for Trajectory-based HCI Tasks," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, Atlanta, Georgia, United States, 1997, pp. 295-302.
- [5] J. Accot and S. Zhai, "Performance Evaluation of Input Devices in Trajectory-Based Tasks: An Application of the Steering Law," in *Proceedings of the SIGCHI conference on Human factors in computing systems: the CHI is the limit*, Pittsburgh, Pennsylvania, United States, 1999, pp. 466-472.
- [6] J. Accot and S. Zhai, "Scale Effects in Steering Law Tasks," in *Proceedings of the SIGCHI conference on Human factors in computing systems*, Seattle, Washington, United States, 2001, pp. 1-8.
- [7] R. Balakrishnan and I. S. MacKenzie, "Performance Differences in the Fingers, Wrist, and Forearm in Computer Input Control," in *Proceedings of the SIGCHI conference on Human factors in computing systems*, Atlanta, Georgia, United States, 1997, pp. 303-310.

- [8] C. Appert, M. Beaudouin-Lafon, and W. E. Mackay, "Context Matters: Evaluating Interaction Techniques with the CIS Model," in *Proceedings of the HCI'04 Conference on People and Computers XVIII*, Leeds, United Kingdom, 2004, pp. 279-296.
- [9] N. E. Wiseman, H. U. Lemke, and J. O. Hiles, "PIXIE: A New Approach to Graphical Man-Machine Communications," in *Proceedings of the 1969 CAD Conference*, Southampton, England, 1969, p. 463.
- [10] D. Hopkins, "The Design and Implementation of Pie Menus," *Dr. Dobb's Journal*, vol. 16, pp. 16-26, 1991.
- [11] J. Callahan, D. Hopkins, M. Weiser, and B. Shneiderman, "An Empirical Comparison of Pie vs. Linear Menus," in *Proceedings of the SIGCHI conference on Human factors in computing systems*, Washington, D.C., United States, 1988, pp. 95-100.
- [12] Trolltech, "Pie Menu," Trolltech. 2008. [Online]. Available: <http://troll.no/products/qt/addons/catalog/4/Widgets/qtpiemenu>. [Accessed: 5/21/2008].
- [13] P. J. Lyons, M. Pitchforth, D. Page, T. Given, and M. D. Apperley, "The Oval Menu - Evolution and Evaluation of a Widget," in *6th Australian Conference on Computer-Human Interaction*, New Zealand, 1996, pp. 252-259.
- [14] L. C. Anderson, K. L. Holman, and M. G. B. F. Van Swaaij. (1998). United States Patent No. 5,828,360. USPTO.
- [15] S. R. Abrams, D. V. Oppenheim, D. P. Pazel, and J. L. Wright. (2001). United States Patent No. 6,285,367. USPTO.
- [16] R. G. Atkinson. (1997). United States Patent No. 5,701,424. USPTO.
- [17] S. W. Driskell. (1997). United States Patent No. 5,596,699. USPTO.
- [18] E. J. Selker. (2003). United States Patent No. 6,549,219. USPTO.

- [19] M. Vayda and N. Weyl. (1998). United States Patent No. 5,798,760. USPTO.
- [20] K. Hinckley, P. Baudisch, G. Ramos, and F. Guimbretiere, "Design and Analysis of Delimiters for Selection-Action Pen Gesture Phrases in Scriboli," in *Proceedings of the SIGCHI conference on Human factors in computing systems*, Portland, Oregon, United States, 2005, pp. 451-460.
- [21] D. Venolia and F. Neiberg, "T-Cube: A Fast, Self-Disclosing Pen-Based Alphabet," in *Proceedings of the SIGCHI conference on Human factors in computing systems: celebrating interdependence*, Boston, Massachusetts, United States, 1994, pp. 265-270.
- [22] G. Kurtenbach, G. W. Fitzmaurice, R. N. Owen, and T. Baudel, "The Hotbox: Efficient Access to a Large Number of Menu-Items," in *Proceedings of the SIGCHI conference on Human factors in computing systems: the CHI is the limit*, Pittsburgh, Pennsylvania, United States, 1999, pp. 231-237.
- [23] D. Hopkins, "Pie Menu Cookbook," 1987. [Online]. Available: <http://www.donhopkins.com/drupal/node/94>. [Accessed: 9/16/2007].
- [24] J. Accot and S. Zhai, "More than Dotting the I's --- Foundations for Crossing-based Interfaces," in *Proceedings of the SIGCHI conference on Human factors in computing systems: Changing our world, changing ourselves*, Minneapolis, Minnesota, United States, 2002, pp. 73-80.
- [25] T. Winograd and F. Guimbreti re, "Visual Instruments for an Interactive Mural," in *CHI '99 Extended Abstracts on Human Factors in Computing Systems*, Pittsburgh, Pennsylvania, United States, 1999, pp. 234-235.
- [26] A. Cockburn and A. Firth, "Improving the Acquisition of Small Targets," in *People and Computers XVII: British Computer Society Conference on Human Computer Interaction*, Bath, England, 2003, pp. 181-196.
- [27] G. Lapizco-Encinas and A. Rodr guez, "CrossEd: Novel Interaction for Pen-Based Systems," University of Maryland, College Park, Maryland, United States,

- [28] G. Apitz and F. Guimbretière, "CrossY: A Crossing-Based Drawing Application," in *Proceedings of the 17th annual ACM symposium on User interface software and technology*, Santa Fe, New Mexico, United States, 2004, pp. 3-12.
- [29] F. Guimbretière and T. Winograd, "FlowMenu: Combining Command, Text, and Data Entry," in *Proceedings of the 13th annual ACM symposium on User interface software and technology*, San Diego, California, United States, 2000, pp. 213-216.
- [30] B. B. Bederson, "Interfaces for Staying in the Flow," in *Ubiquity*. Vol. 5, 2004, p. 1.
- [31] S. Pook, E. Lecolinet, G. Vaysseix, and E. Barillot, "Control Menus: Execution and Control in a Single Interactor," in *CHI '00 extended abstracts on Human factors in computing systems*, The Hague, The Netherlands, 2000, pp. 263-264.
- [32] G. Kurtenbach and W. Buxton, "Issues in Combining Marking and Direct Manipulation Techniques," in *Proceedings of the 4th annual ACM symposium on User interface software and technology*, Hilton Head, South Carolina, United States, 1991, pp. 137-144.
- [33] G. P. Kurtenbach, A. J. Sellen, and W. A. S. Buxton, "Some Articulatory and Cognitive Aspects of "Marking Menus": an Empirical Study," Toronto, Canada: Computer Systems Research Institute, University of Toronto, 1991, p. 31.
- [34] G. Kurtenbach, "The Design and Evaluation of Marking Menus," Ph.D. Thesis, University of Toronto, Toronto, Canada, 1993.
- [35] M. A. Tapia and G. Kurtenbach, "Some Design Refinements and Principles on the Appearance and Behavior of Marking Menus," in *Proceedings of the 8th annual ACM symposium on User interface and software technology*, Pittsburgh, Pennsylvania, United States, 1995, pp. 189-195.
- [36] G. Kurtenbach and W. Buxton, "User Learning and Performance with Marking Menus," in *Proceedings of the SIGCHI conference on Human factors in computing systems: celebrating interdependence*, Boston, Massachusetts, United States, 1994, pp. 258-264.

- [37] G. P. Kurtenbach. (1997). United States Patent No. 5,689,667. USPTO.
- [38] S. Zhao and R. Balakrishnan, "Simple vs. Compound Mark Hierarchical Marking Menus," in *Proceedings of the 17th annual ACM symposium on User interface software and technology*, Santa Fe, New Mexico, United States, 2004, pp. 33-42.
- [39] W. E. Mackay, "Which Interaction Technique Works When? Floating Palettes, Marking Menus and Toolglasses support different task strategies," in *Proc. Conference on Advanced Visual Interfaces, AVI'02*, Trento, Italy, 2002, pp. 203-208.
- [40] M. McGuffin, N. Burtnyk, and G. Kurtenbach, "FaST Sliders: Integrating Marking Menus and the Adjustment of Continuous Values," in *Proceedings of Graphics Interface 2002*, Calgary, Alberta, Canada, 2002, pp. 35-42.
- [41] G. M. Smith and M. C. Schraefel, "The Radial Scroll Tool: Scrolling Support for Stylus or Touch-Based Document Navigation," in *Proceedings of the 17th annual ACM symposium on User interface software and technology*, Santa Fe, New Mexico, United States, 2004, pp. 53-56.
- [42] Microsoft, "Tablet PC Support," Microsoft. 2007. [Online]. Available: <http://www.microsoft.com/windows/products/windowsvista/features/details/tabletpc.mspx>. [Accessed: 6/17/2007].
- [43] Mozilla, "Mouse Gestures," Mozilla. 2000. [Online]. Available: <http://optimoz.mozdev.org/gestures/>. [Accessed: 6/17/2007].
- [44] D. Hopkins, "Why Pie Menus Aren't Ubiquitous?," Inf@Vis! 2003. [Online]. Available: <http://www.infovis.net/printMag.php?num=125&lang=2>. [Accessed: 6/17/2007].
- [45] Linden Labs. (2007). Second Life (Version 1.18.2).
- [46] Mangold International. (2007). LogSquare (Version 3.0). Arnstorf: Mangold International.

- [47] F. E. Ritter, L. J. Schooler, J. S. Neil, and B. B. Paul, "Learning Curve, The," in *International Encyclopedia of the Social & Behavioral Sciences* Oxford: Pergamon, 2001, pp. 8602-8605.
- [48] D. Zambonini, "Plotting the exact X/Y coordinates of clicks on a page," O'Reilly. 2006. [Online]. Available: http://www.oreillynet.com/xml/blog/2006/02/plotting_the_exact_xy_coordina.html. [Accessed: 6/22/2008].
- [49] I. S. MacKenzie, A. Sellen, and W. A. S. Buxton, "A Comparison of Input Devices in Element Pointing and Dragging Tasks," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, New Orleans, Louisiana, United States, 1991, pp. 161 - 166.
- [50] SPSS. (2007). SPSS (Version SPSS Graduate Student Version for Mac Release 16.0.1). Chicago: SPSS Inc.
- [51] J. Boritz, K. S. Booth, and W. B. Cowan, "Fitts's law studies of directional mouse movement.," in *Graphics Interface '91*, Calgary, Alberta, Canada, 1991, pp. 216-223.

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APPENDIX

Test Script Used in the Experiment

Before the subject arrives figure out the next file name (they are intended to be run in sequential order to preserve the blocking of the variables).

Write the filename here: trial_____.cvs

Launch LogSquare Recorder (icon on desktop)

Minimize LogSquare Recorder

Launch Second Life® (icon on desktop) and select "Connect"

Minimize Second Life®

Wipe mouse with a wipe

After the subject arrives start the script:

Hi, I'm Brent and I have been asked to be an impartial administrator for this experiment. You may have noticed, and think it a bit odd, that I am reading from a script. This is to make sure everyone receives the same instructions.

First I need to get your signature on a consent form. There are guidelines for using humans as experimental subjects, and this form explains how the experiment affects you, and gets your permission to use you as a subject. It also helps make clear that your participation is totally voluntary. You can decide not to take part, and may drop out at any point, and there will be no repercussions. Please look over this form and sign it if you agree.

Get signed form

Would you like a copy for yourself?

If yes, give them a blank copy

We're here to test a new way of interacting with a computer. I would like to make it clear from the start that this not a test of your abilities, it is strictly the feature I am concerned with. You may ask questions at any time, but I may not answer, since I want to know how a user would act independently.

The system in front of you is already set-up with the necessary software. I'd like you to arrange the computer, mouse pad and mouse, and adjust the chair to your liking.

Give them time to adjust.

Double check that they are comfortable.

As an introduction, let's look at some menus you are probably familiar with. You probably know that if you right click on an item on the screen a menu appears. Currently the desktop is showing on the computer. Please choose an item on the desktop and right click on it to bring up the menu.

You'll notice a menu has appeared that has a list of functions you can perform on the item you right clicked on. This is called a linear menu, since the items are listed in a linear top- to- bottom order list.

Choose the "Properties" item in the list. You'll notice a dialog appears that shows the properties of the item you originally selected. Dismiss this dialog by pressing cancel or the close box.

Now let's look at a different type of menu. Choose "Second Life" from the running programs at the bottom of the screen. You'll see an avatar standing near some chairs. Right click on one of the chairs.

You'll notice a menu has appeared. This is a Pie Menu. Similar to the linear menu, it has items that you can select. Choose the "Sit" item. You'll notice the avatar sat down. Let's try that one more time. Click the "Stand Up" button along the bottom. Now choose a chair and right click on it. Choose the "Sit" item. Have you used this type of menu before?

Circle response: yes/no

Note their handedness (in which hand are they holding the mouse)

Circle one: left/right

Also note their gender. Circle one: male/female

Those are two types of menus in existing software. I want to try a new type of menu. Let me get the test software ready.

Quit Second Life®

Launch the test app (icon on desktop)

Choose the file with the trial name written above

Choose LogSquare in the tray and start recording

The program you see here is the test application we will be using. Before you begin, let me explain how this test will work. As you can see there are instructions in the middle of the screen. The instructions also serve as the item of interest – or - the item you right click on, just like you right clicked on the icon on the desktop to display the linear menu, or the chair to display the pie menu. The instructions tell you what to choose from the menu once it appears. Once I tell you to start, read the instructions then select what you are instructed to as quickly and accurately as possible.

Once you have made a selection you will be instructed to select a new item. This will continue for a number of trials. There will be 24 trials. Please continue to follow the instructions and select items as quickly and accurately as possible. If you think you made a mistake, simply ignore it and move on. After 24 trials a dialog will appear to notify you that a new test set is about to begin. This is a good time to pause or take a break. The new set will have something different about the menus. There are a total of ten test sets, each with 24 trials, and you will be notified before each set begins.

I'll stay for the first few test sets to make sure you're doing OK, but I may leave after that.

Do you have any questions?

Don't answer questions about the menus themselves, like "what exactly am I testing". Do answer questions about length (less than ½ hour), repeat information about the number of trials, sets, times, break, or even repeat the sentence above with the dashed underline.

You may begin.

Resist answering questions at this point. Explain that you want to see what a user would do if help wasn't available.

If they struggle with the first step (they don't right click) for more than 30 seconds, reread the sentence above with the dashed underline.

If after 1 minute they still haven't right clicked on the instructions, tell them to right click on the instructions.

If they ask "what do I do now?" after they first activate the menu, do not answer. Explain that you want to see what a user would do if help wasn't available. After 30 seconds tell them to "follow what the instructions tell you". After at least a minute if it looks hopeless thank them and abort the test.

When they are done:

All done? Good. Let me stop the software.

Bring up LogSquare recorder from the tray and stop recording.

Thank you so much for participating. The information from the experiment is very valuable. Please don't discuss the experiment with other employees, because we don't want them to be biased in any way by what you might say. Also, please don't discuss it with any customers since it is still just an idea and is confidential.

If the test was not completed – make a note and get help deleting the files (the test will need to be reused to guarantee proper variable blocking)

Notes:
